

Satellite Attitude Determination and Control – Introduction

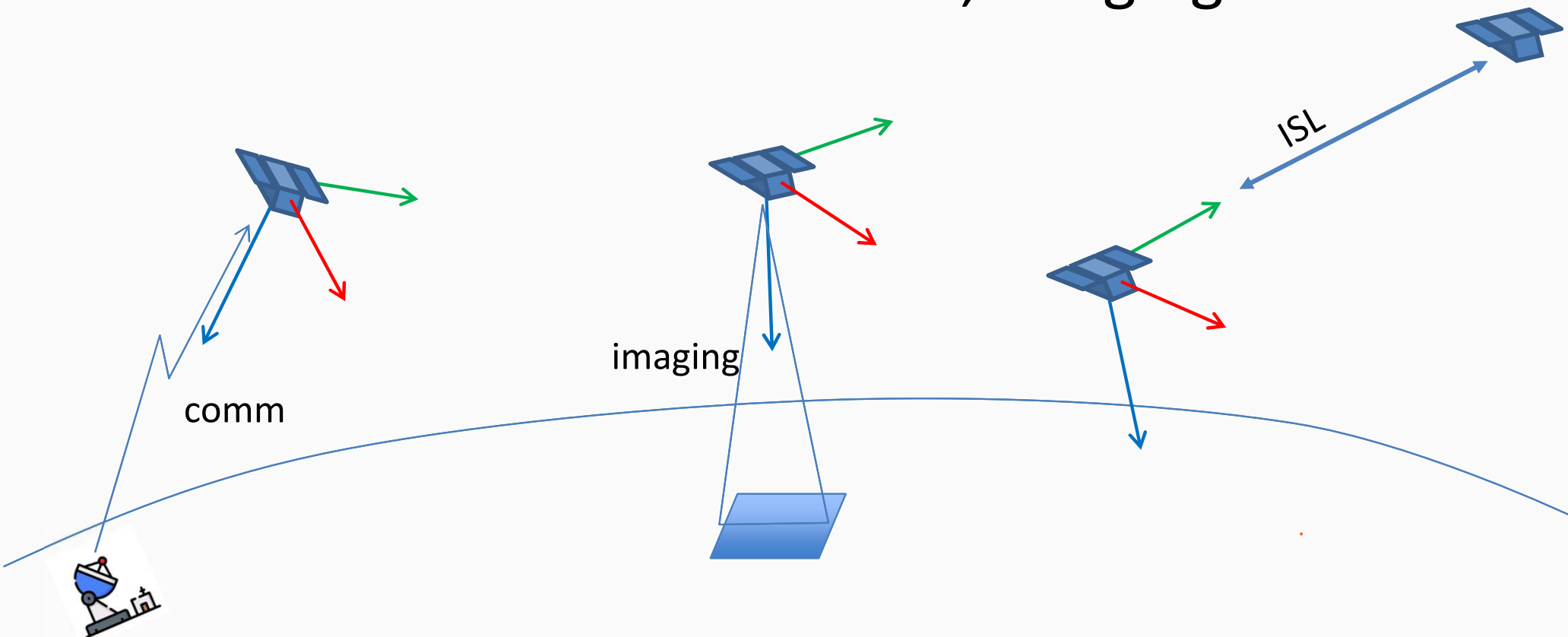
Dr. N. Nagarajan

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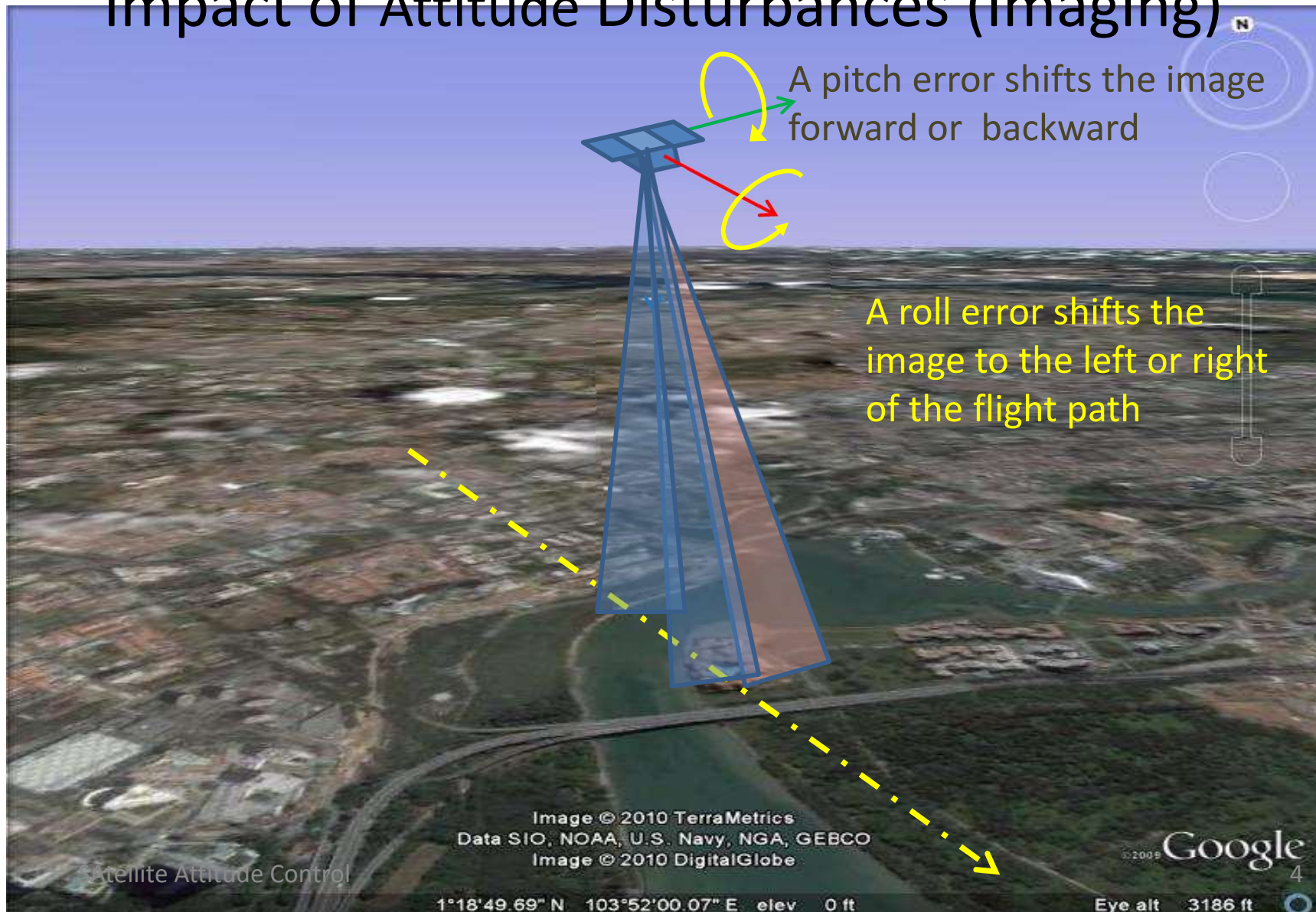
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Why Attitude is needed

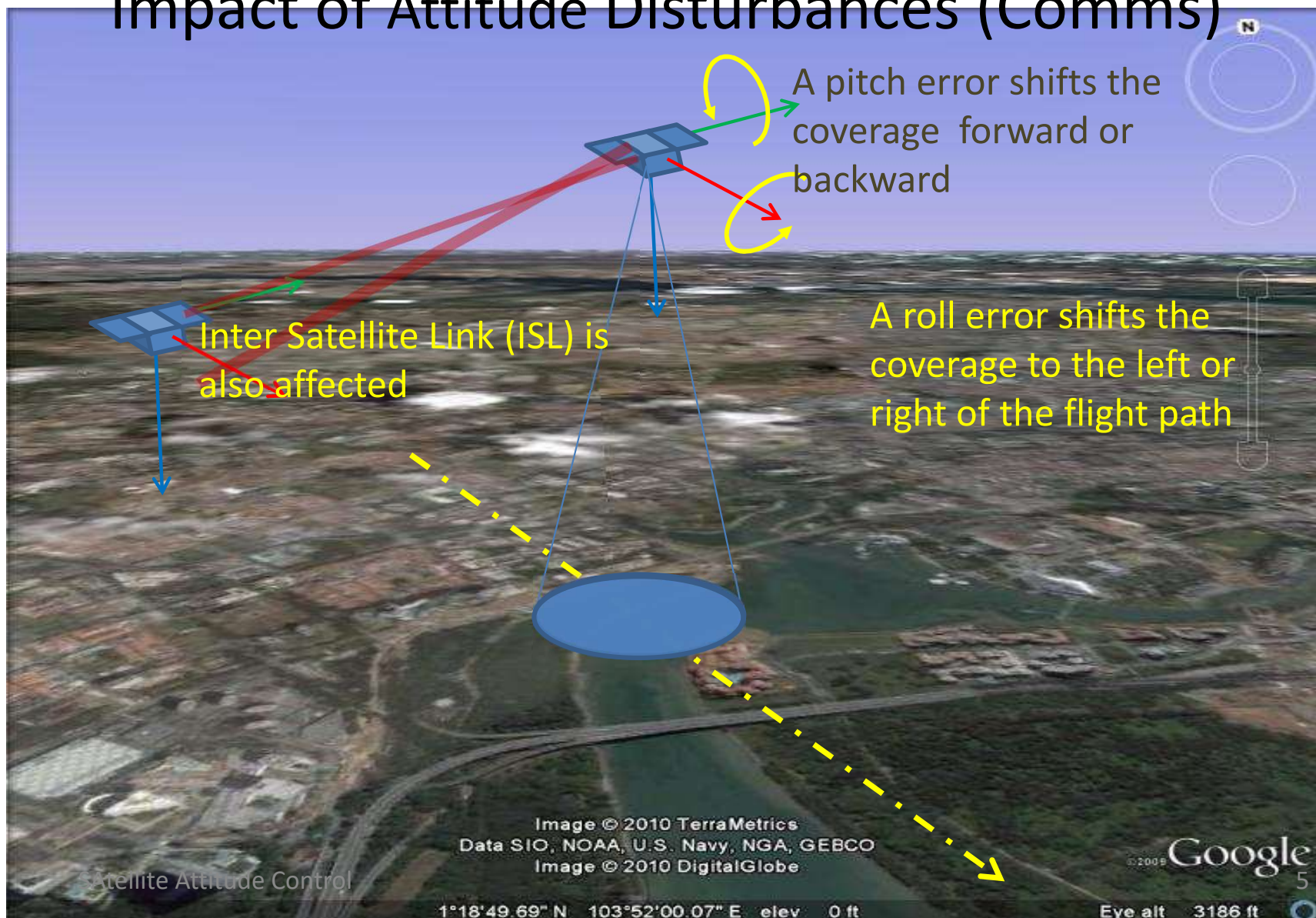
- Attitude defines where antennas, sensors, cameras and instruments point.
- Essential for communication, imaging



Impact of Attitude Disturbances (Imaging)



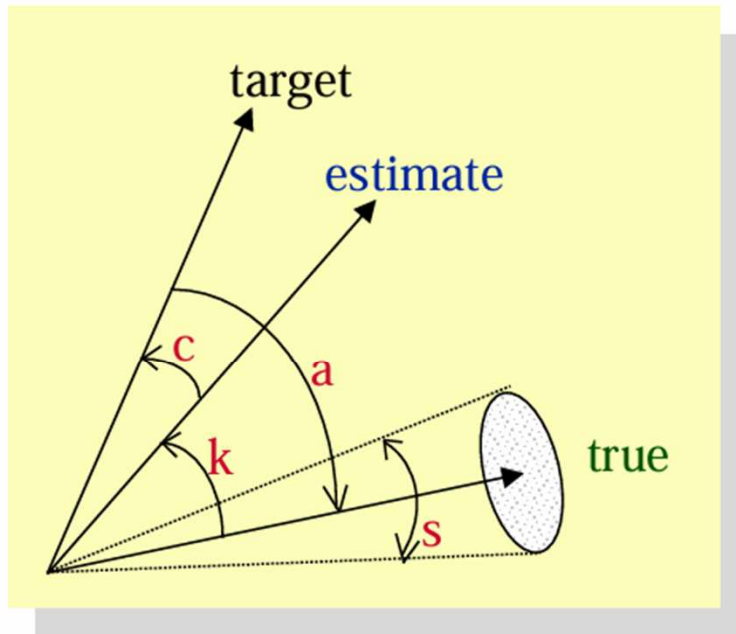
Impact of Attitude Disturbances (Comms)



Orbit Vs Attitude

- Orbit defined by (Pos, Vel) OR Kepler Elements
- Orbit defines the motion of a satellite as a point mass
- $F=ma$ for orbit motion
- Perturbations: Aerodynamic drag, SRP, Earth Oblateness, Third Body, Fuel leakage, Thruster misalignment
- Attitude defined by Euler Angles, rates
- Attitude defines rotation of the body about its center of mass
- $T = I\dot{\omega}$ for attitude rotation
- Perturbations: Aerodynamic torque, SRP, Residual magnetic dipole, Thruster misalignment

Attitude Pointing Definitions



target	desired pointing direction
true	actual pointing direction (mean)
estimate	estimate of true (instantaneous)
a	pointing accuracy (long-term)
s	stability (peak-peak motion)
k	knowledge error
c	control error

a = pointing accuracy = attitude error
s = stability = attitude jitter

Attitude Pointing Definition (MIT Open Courseware)

Source:
G. Mosier
NASA GSFC

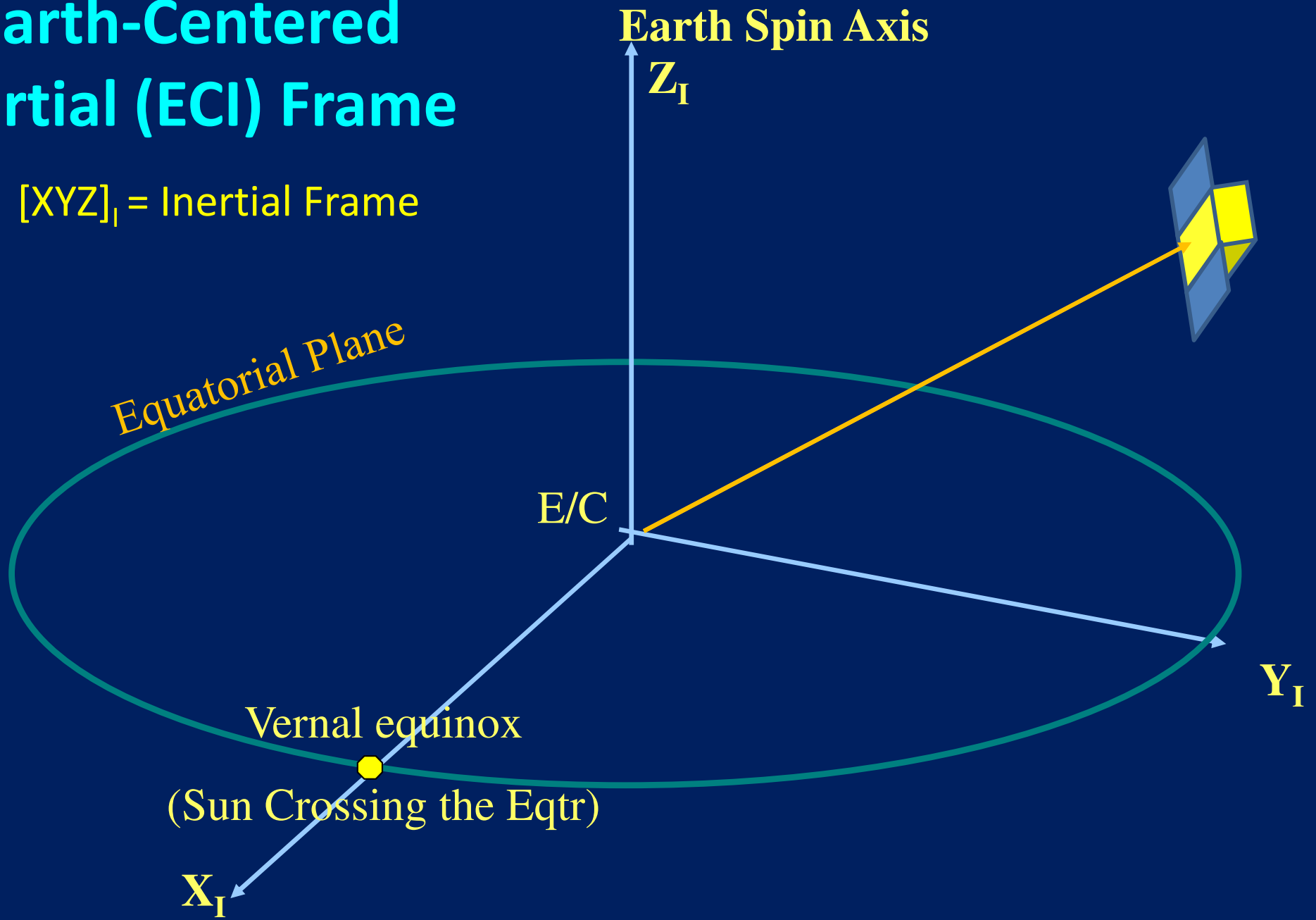
REFERENCE FRAMES

Reference Frames

- Why frames are needed.
 - Reference frames are needed to specify the orientation of the body axes, how they with time.
 - To calculate the torques and integrate the attitude motion
 - To transform the measurements from different sensors to a common frame.
- Earth-Centered Inertial (ECI).
- Earth-Centered Earth Fixed (ECEF).
- Orbit Reference Frame.
- Body frame of the satellite.
- Sensor Frame

Earth-Centered Inertial (ECI) Frame

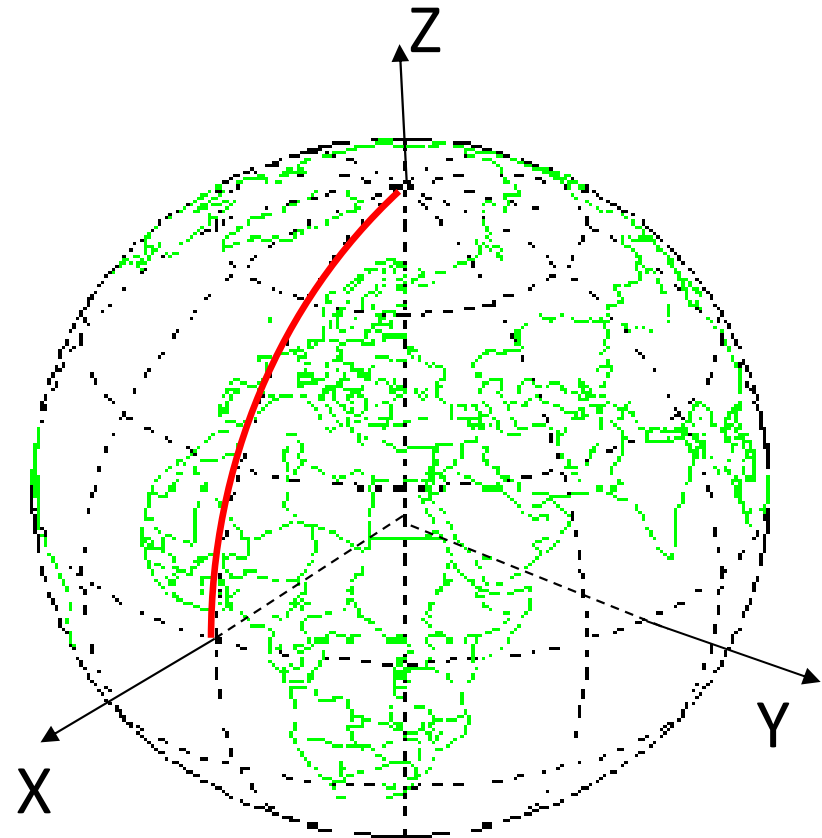
$[XYZ]_I$ = Inertial Frame



Earth Fixed Coordinate System

$[XYZ]_g$

- Z along spin axis
- X along the equator on the Greenwich meridian
- $Y = Z \times X$
- Useful to define the location of ground station, direction of geomagnetic field...
- X, Y, Z components are used to calculate the longitude, latitude and altitude.

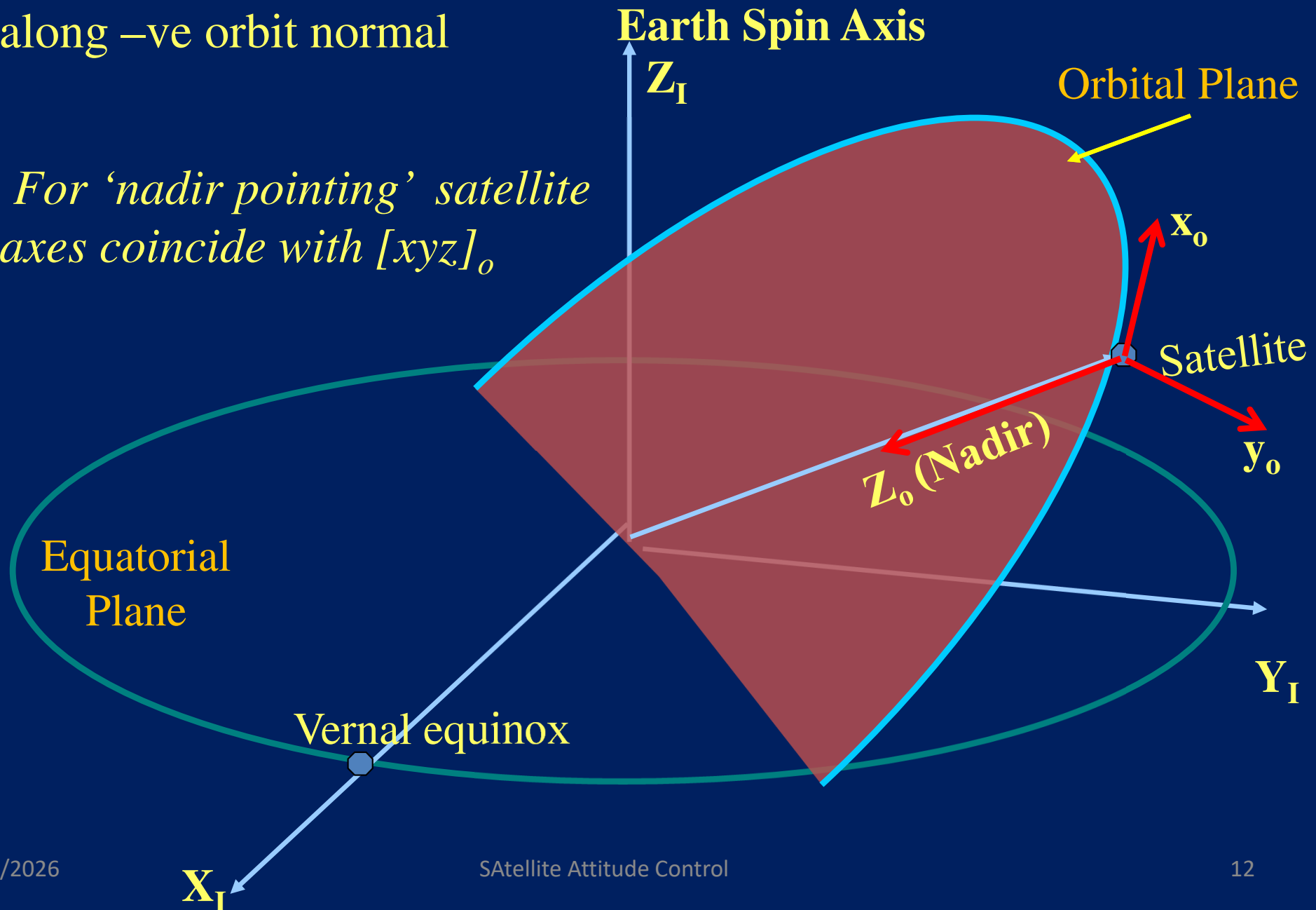


Orbit Reference Frame $[XYZ]_o$

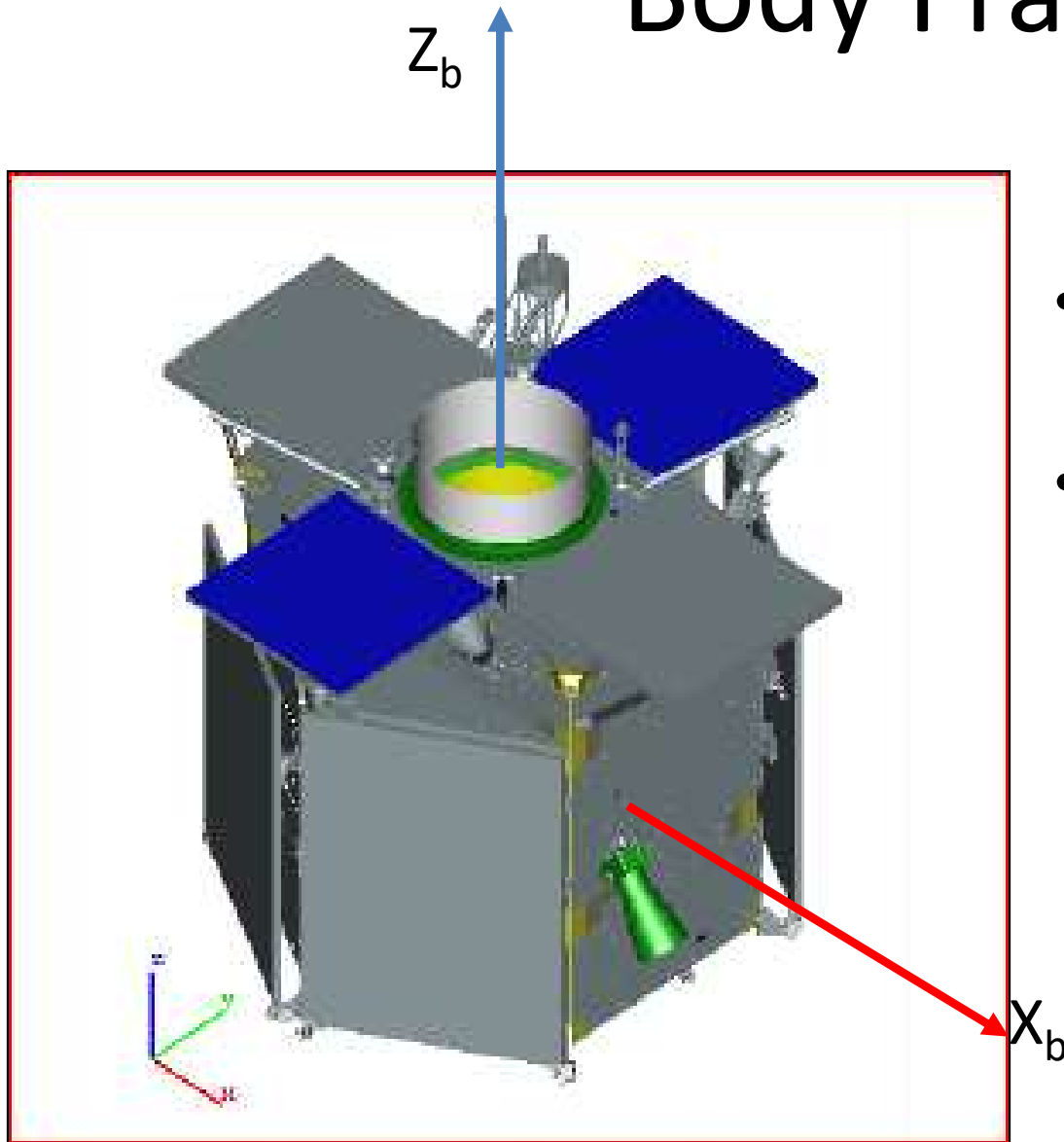
z_o = along -ve radius vector

y_o = along -ve orbit normal

Note: For 'nadir pointing' satellite body axes coincide with $[xyz]_o$



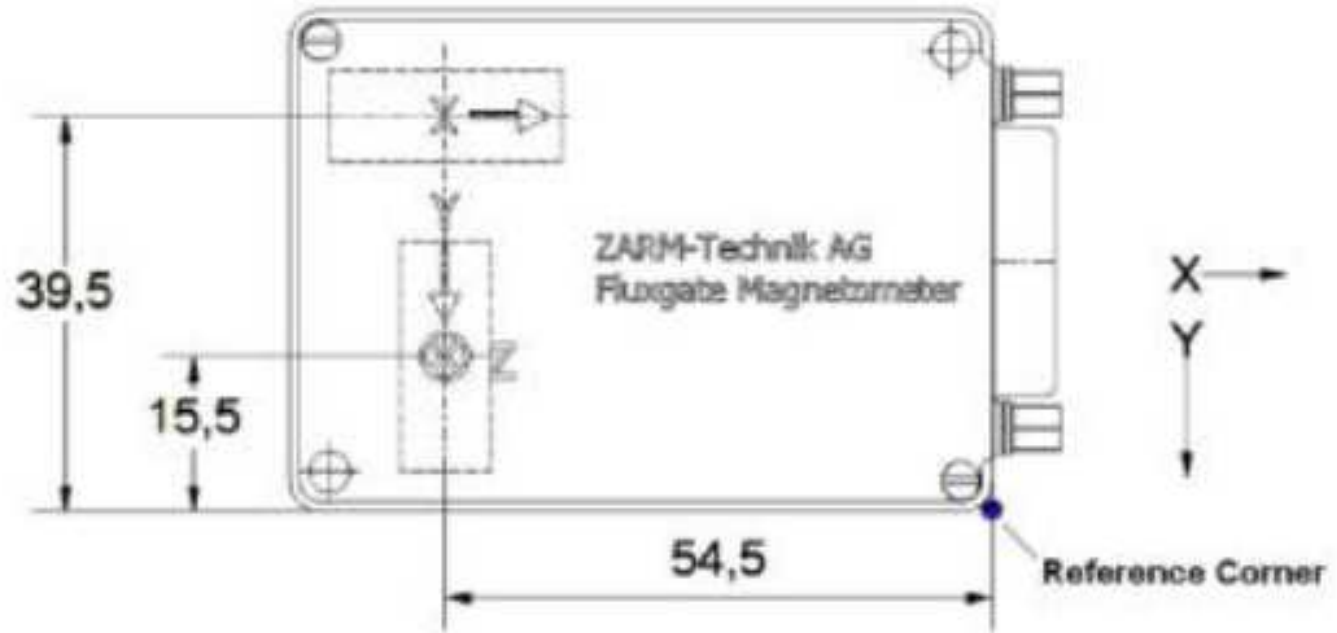
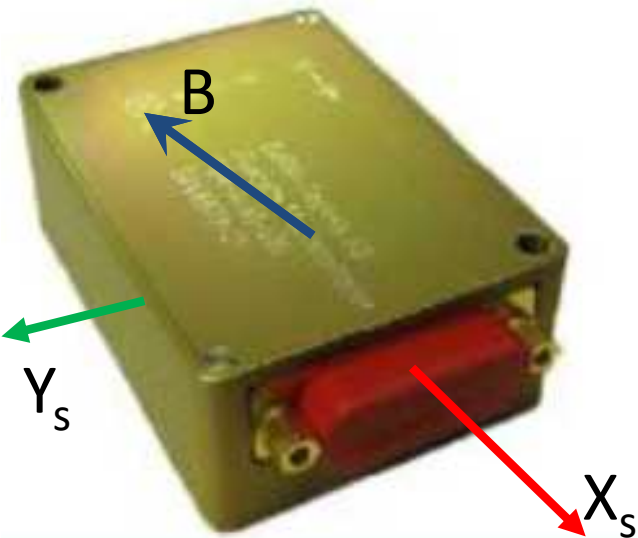
Body Frame



- This is useful to define the mechanical configuration
- Mounting of sensors, actuators and other modules are specified in this system

Sensor Fixed Coordinate System

Magnetic sensor



- This is useful to define the output of the sensor
- The orientation of the sensor frame with ref to the body frame define the mounting of the sensors

COORDINATE ROTATIONS

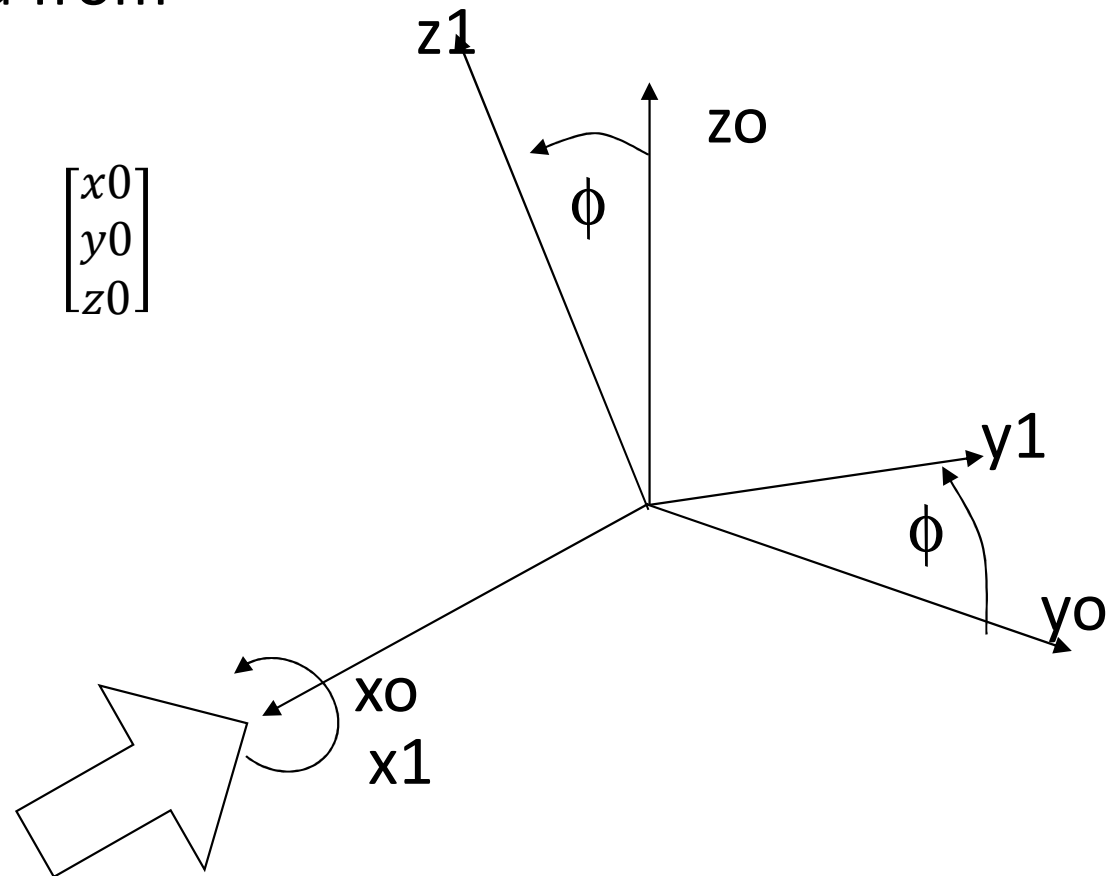
Coordinate Rotations

- What for ?
 - Coordinate rotations are required to transform a vector from a given reference frame to another frame.
 - Measurements from sensor axes need to be transformed to body axes and then to Inertial Frame for estimating body attitude
 - During simulation, the Sun, Magnetic, Star Vectors need to be transformed from Inertial to sensor frame to get sensor measurements
- This may involve simple rotation about one axis or complex rotations involving a series of rotations about different axes.

Simple Rotations (about x axis)

A rotation is positive, if the rotation is anti-clockwise when viewed from the positive x- axis

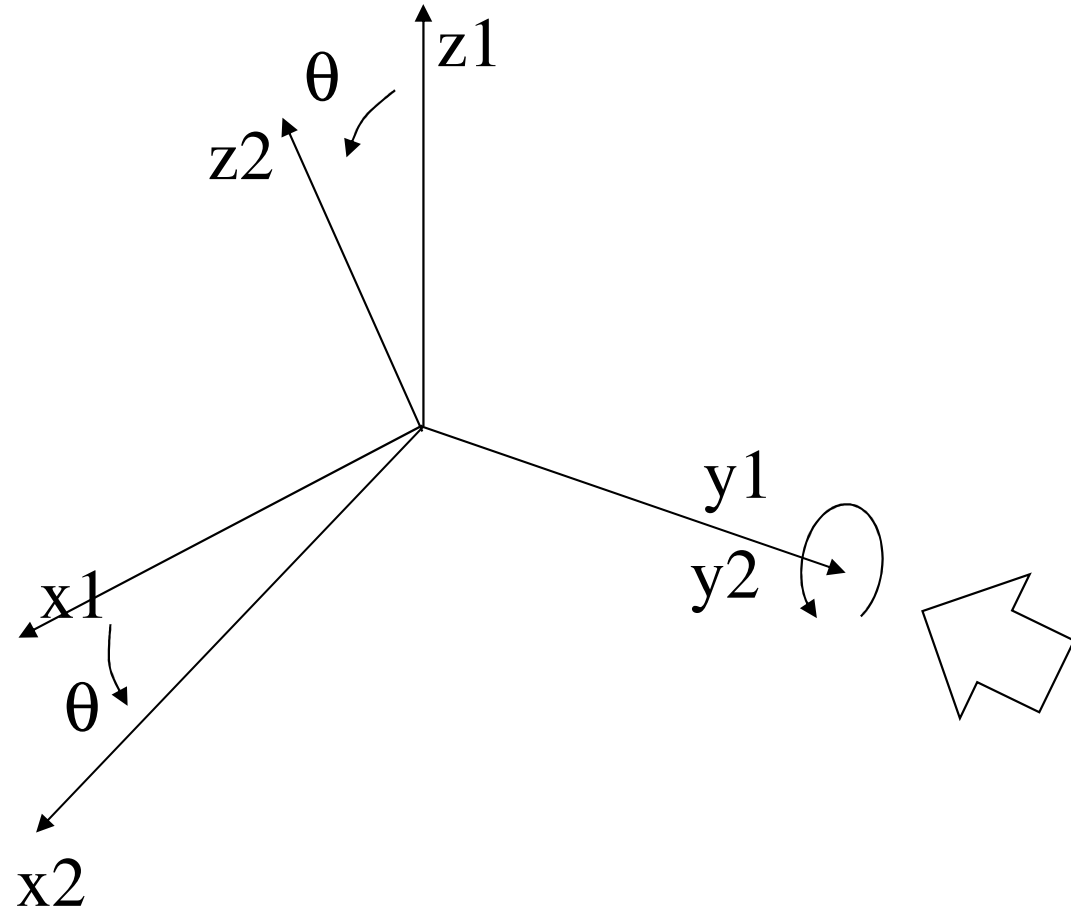
$$\begin{bmatrix} x1 \\ y1 \\ z1 \end{bmatrix} = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{pmatrix}}_{R1} \begin{bmatrix} x0 \\ y0 \\ z0 \end{bmatrix}$$



Simple Rotations (about y axis)

A rotation is positive, if the rotation is anti-clockwise when viewed from the positive y- axis

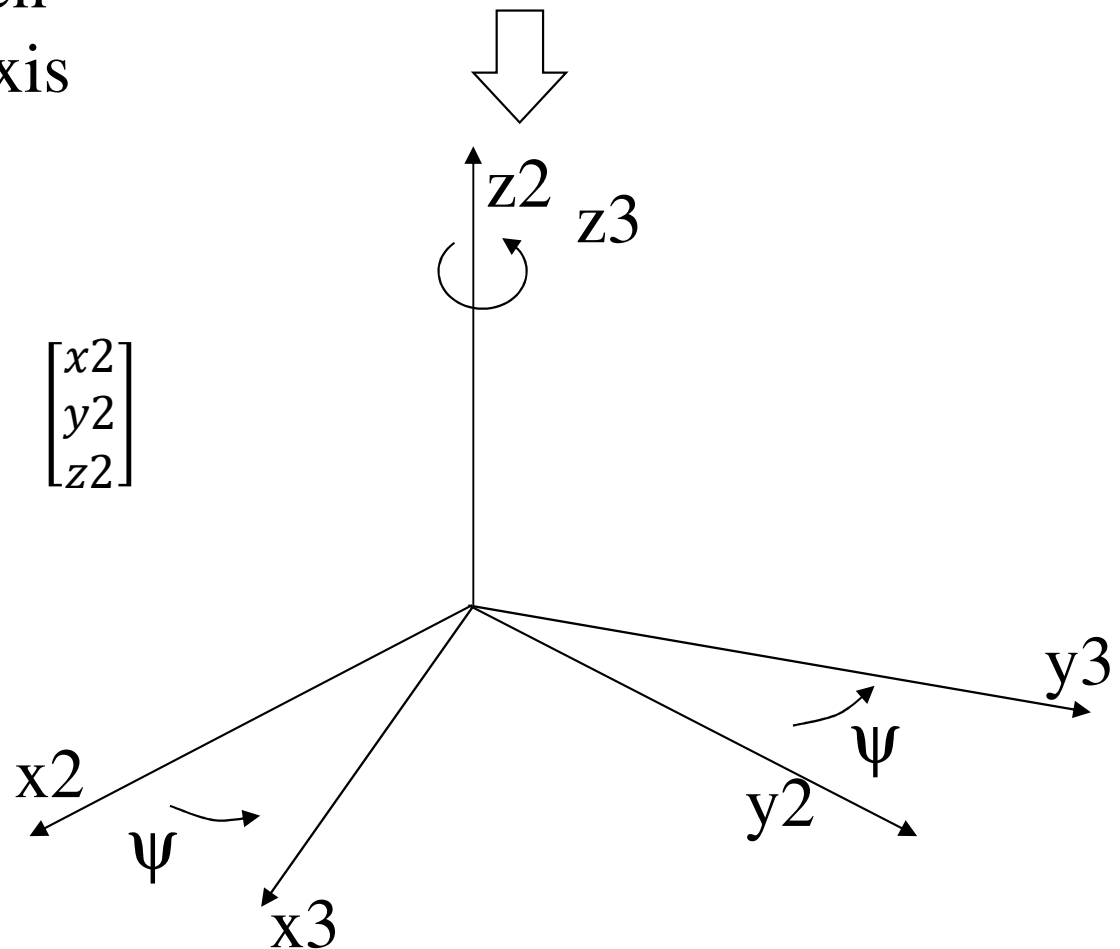
$$\begin{bmatrix} x2 \\ y2 \\ z2 \end{bmatrix} = \underbrace{\begin{pmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{pmatrix}}_{R2} \begin{bmatrix} x1 \\ y1 \\ z1 \end{bmatrix}$$



Simple Rotations (about z axis)

A rotation is positive, if the rotation is anti-clockwise when viewed from the positive z- axis

$$\begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} = \underbrace{\begin{pmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{R_3} \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$$



Multiple rotations

Even complex multiple rotations, can be done as simple sequential rotations. The order is very important. Some typical rotations.

Type	1 st rotation	2 nd rotation	3 rd rotation
1-2-3	ϕ about x_0	θ about y_1	ψ about z_2
3-1-3	ϕ about z_0	θ about x_1	ψ about z_2
3-2-3	ϕ about z_0	θ about y_1	ψ about z_2

$$\begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} = R_3 \cdot R_2 \cdot R_1 \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix}$$

ATTITUDE REPRESENTATION

Attitude Representation Methods

- Euler angles (Roll, Pitch, Yaw).
- Direction cosine matrices.
- Quaternions

Attitude Definition (Euler Angles)



Typical Transformation Matrix (C) using Euler Angles

- For the Euler rotation sequence 1-2-3 (roll–pitch–yaw),

$$C = R_z(\psi) \cdot R_y(\theta) \cdot R_x(\phi)$$

$$C_{123} = \begin{bmatrix} \cos \theta \cos \psi & \cos \psi \sin \theta \sin \phi - \sin \psi \cos \phi & \cos \psi \sin \theta \cos \phi + \sin \psi \sin \phi \\ \cos \theta \sin \psi & \sin \psi \sin \theta \sin \phi + \cos \psi \cos \phi & \sin \psi \sin \theta \cos \phi - \cos \psi \sin \phi \\ -\sin \theta & \cos \theta \sin \phi & \cos \theta \cos \phi \end{bmatrix}$$

Transforms a vector from the body frame to the Reference Frame

Attitude Representations Comparison

- The **Euler angles** provide good insight about the orientation; but involves trigonometric terms while handling the dynamical equations.
- **Quaternion form** is easy to handle in dynamical equations of ADCS; It's difficult to visualize the orientation when compared to Euler angle form
- **DCM (Direction Cosine Matrix)** for is very useful, to define the attitude for ADCS modes like 'Sun Tracking' , 'Ground Target Tracking' etc and also useful to transform the attitude from one frame to another

SENSORS AND ACTUATORS

Sensors for Attitude Determination

- Sun sensors.
 - Coarse, Fine, Digital types
- Magnetometers.
 - Fluxgate, MEMS type
- Star trackers
 - Single Optical Head (OH) type
 - Multi Optical Head (Ohs) and Electronic Units (EU)
- Gyros.
 - Mechanical, Fiber Optic Gyros, MEMs Gyros

SUN SENSOR

- Measures the sun direction in body frame

Field of view 140°

Update rate 5 Hz typical

Accuracy $\leq 0.1^\circ$ RMS error over 120° FOV

PHYSICAL CHARACTERISTICS

Dimensions 34 mm x 40 mm x 20 mm

Mass <35 g

Power Active: <30mA [150mW]

INTERFACES

Power supply 5 VDC

Data RS-485 UART

Connector 9-way female Nano-D 9-way socket Micro-D

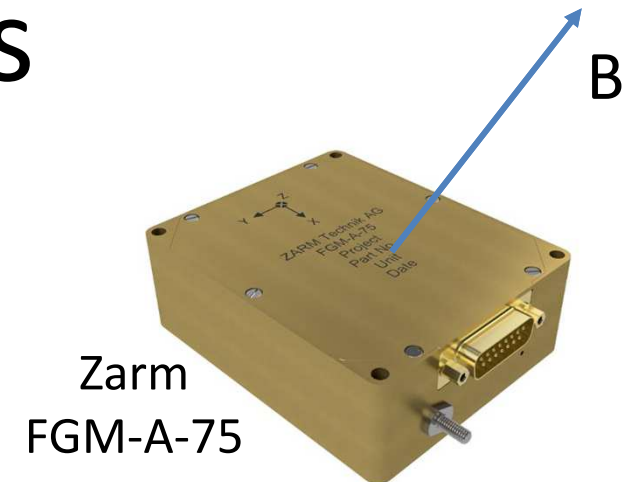


- Multiple sensors are used to acquire the sun from any direction.

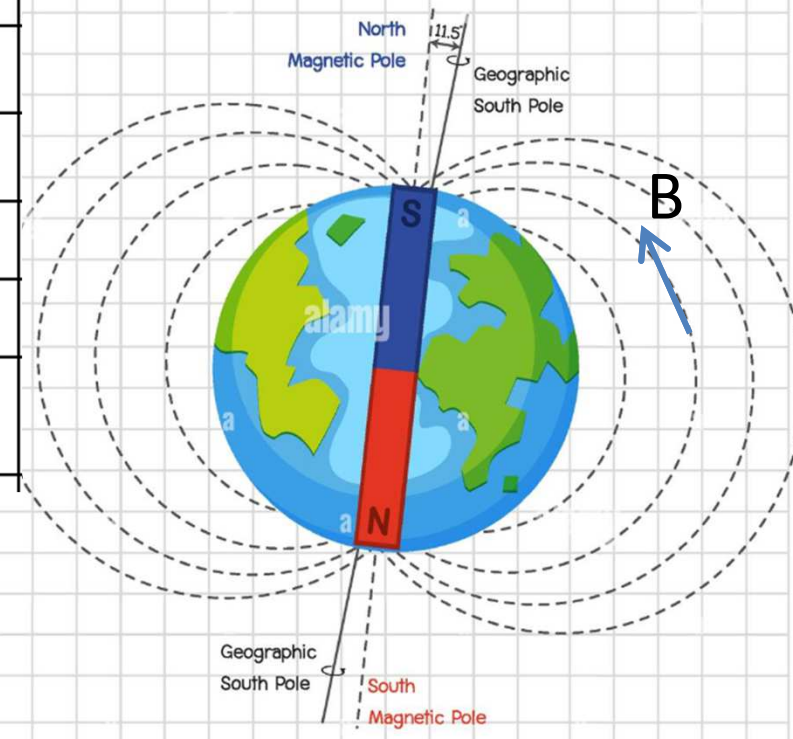
Magnetometers

Measures the magnetic field vector (B)

Parameter	Value
Number of Axes	Three, orthogonal
Sensor type	Analogue fluxgate
Field Measurement Range	$\pm 75\mu\text{T}$
Supply Voltage	$\pm 15\text{V}$
Operating Temperature Range	-30°C to $+60^{\circ}\text{C}$
Non-Operating Temperature Range	-40°C to $+70^{\circ}\text{C}$
Dimensions	100 x 82 x 34mm
Mass	$\leq 330\text{g}$
Connector	DB9 Non-Magnetic male (Space grade)



EARTH'S MAGNETIC FIELD

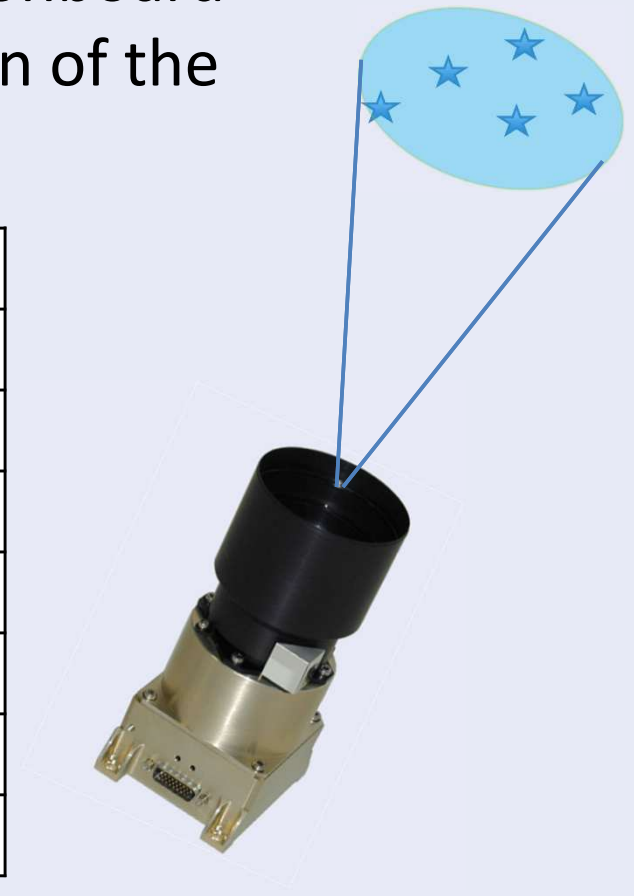


Used for satellite detumbling using torquers.
Used for coarse attitude determinations using Sun
and Magnetometer data

Star Tracker

Images the stars in the FOV, matches with onboard star catalog and determines the orientation of the satellite in inertial space.

Accuracy 2σ (x,y / z axis)	18 arcsec / 122 arcsec
Acquisition probability	> 99.7%
Update rate	4Hz
Field of view	14° x 14°
Time of first acquisition	typ. 800 ms
Power	2.5W
Signal interface	RS422 / RS485
Mass	0.7 to 0.9 kg



Used for precision attitude reference, during imaging, payload data downloading, large angle maneuvers etc

Gyros



3-Axis FoG from Astrofein
Range (per channel) $\pm 40^\circ/\text{s}$
Noise(Random Walk) $\leq 0.1^\circ/\sqrt{\text{h}}$
Mass $< 1\text{Kg}$
Data Interface RS422



1-Axis FoG from Optolink
Range (per channel) $\pm 95^\circ/\text{s}$
Noise(Random Walk) $\leq 0.0007^\circ/\sqrt{\text{h}}$
Mass $< 1\text{Kg}$
Data Interface RS485

Used for measuring body rates precisely. Star tracker attitude, FoG rates together form the precise attitude measurement set ' for imaging, target tracking modes etc

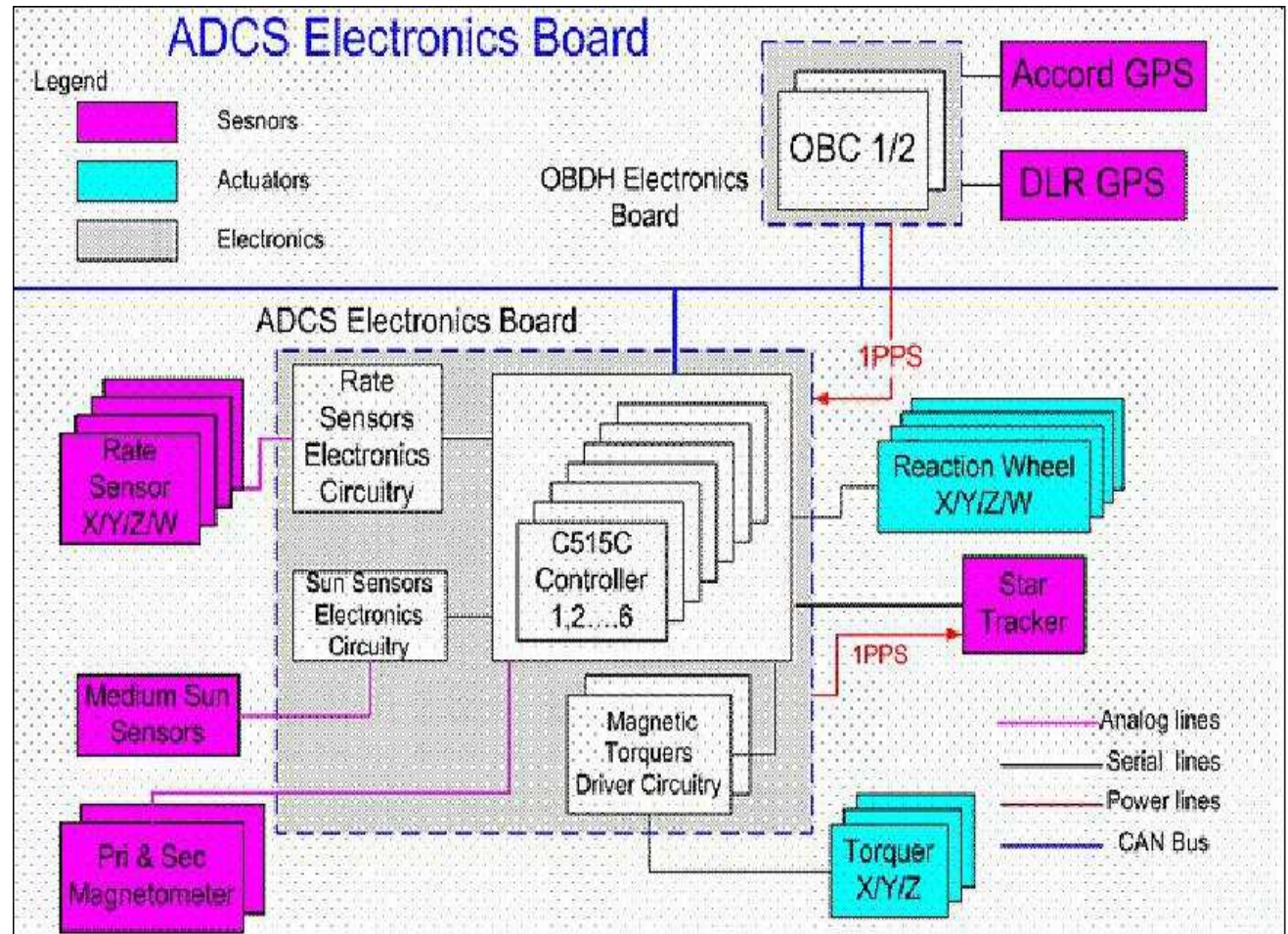
Sensor Fusion

- Multiple sensors are used onboard for
 - Wider coverage around the satellite
 - Sun sensors to acquire the sun from any orientation initially
 - Improved accuracy and steerability
 - multi-head star trackers improves accuracy and satellite can be steered over large angles for payload ops
 - Redundancy
 - To handle failures in space and continue the mission

Sensor Fusion

XSAT ADCS

- 4 Sun Sensors
- 2 Magnetometers
- 4 Rate Sensors
- 4 Reaction wheels
- 3 Torquers (with 2 coils each)
- 2 GPS Rx

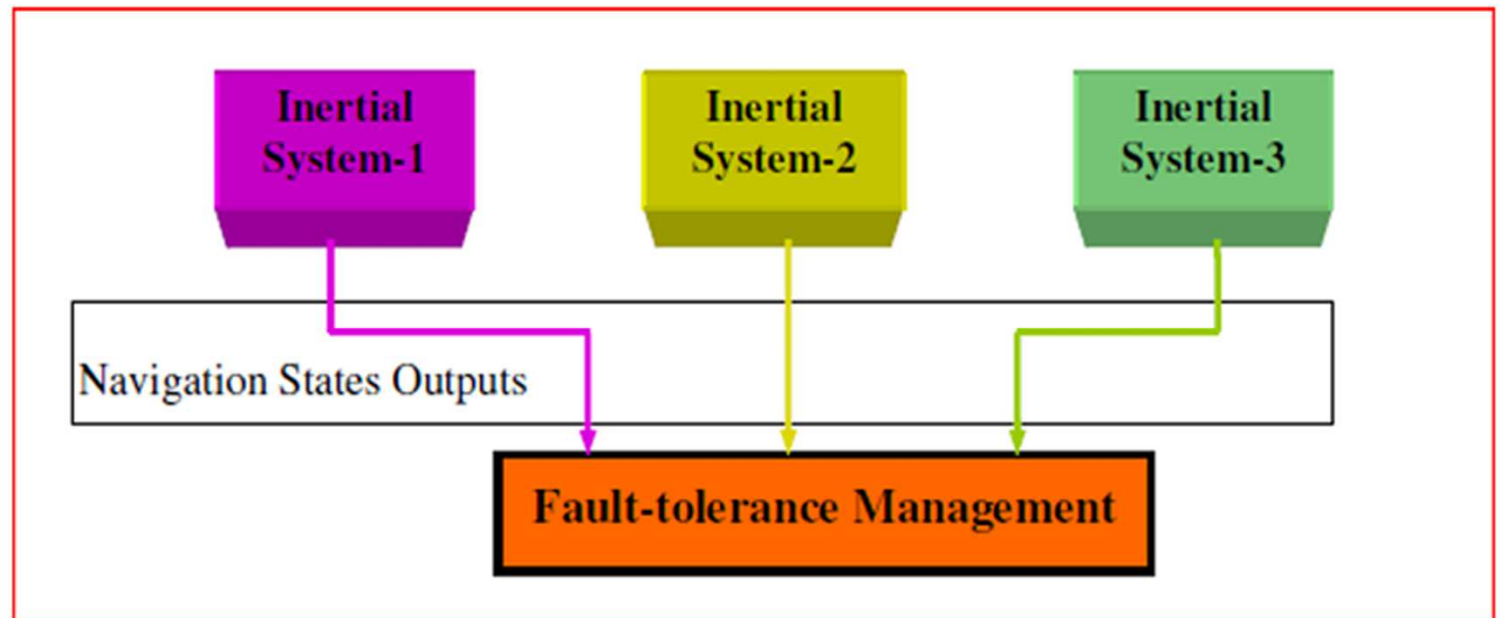


Redundancy & Fault Tolerance

System Level Redundancy

The entire navigation system is a triplex or quadruple System (i.e Sensors, I/F, Processor etc..)

Easy and Simple to implement. Fault Tolerant Management performs consistency checks and failure detection Expensive in terms of cost

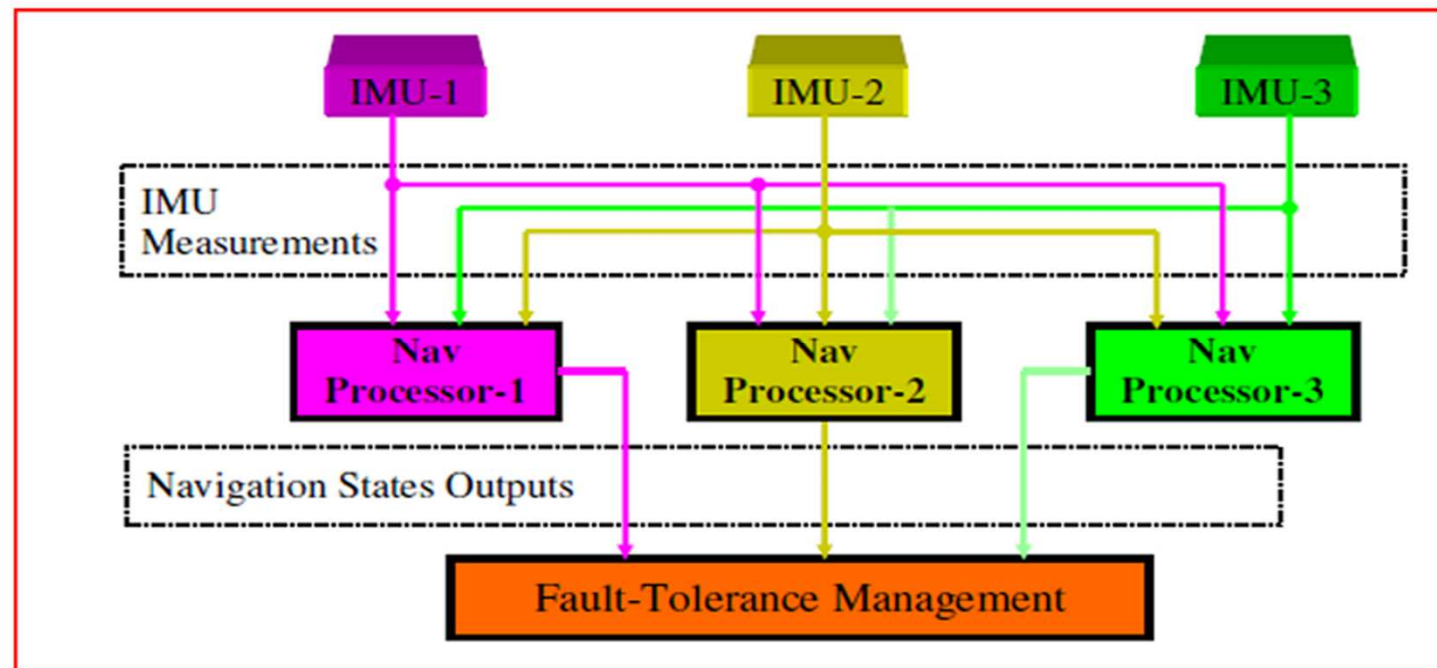


Redundancy & Fault Tolerance

Sensor Level Redundancy

Only the sensor system is a triplex to handle redundancy and fault tolerance

Each navigation processor can use the data of all sensor data

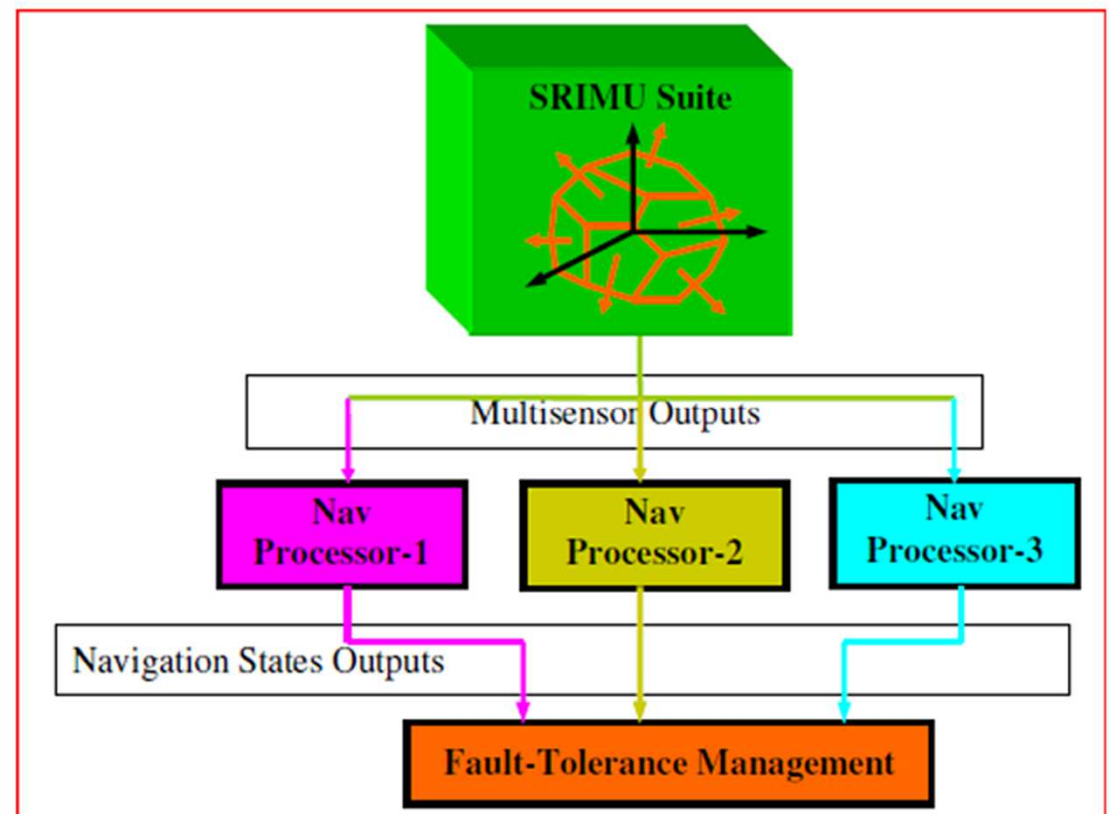


Redundancy & Fault Tolerance

Multi Sensor Redundancy

The sensor system itself, built with redundancy internally (like two orthogonal 3 axis-sensors, with one skewed to handle redundancy and fault tolerance)

Helps to reduce the volume, weight and power requirements.
Thus cost effective

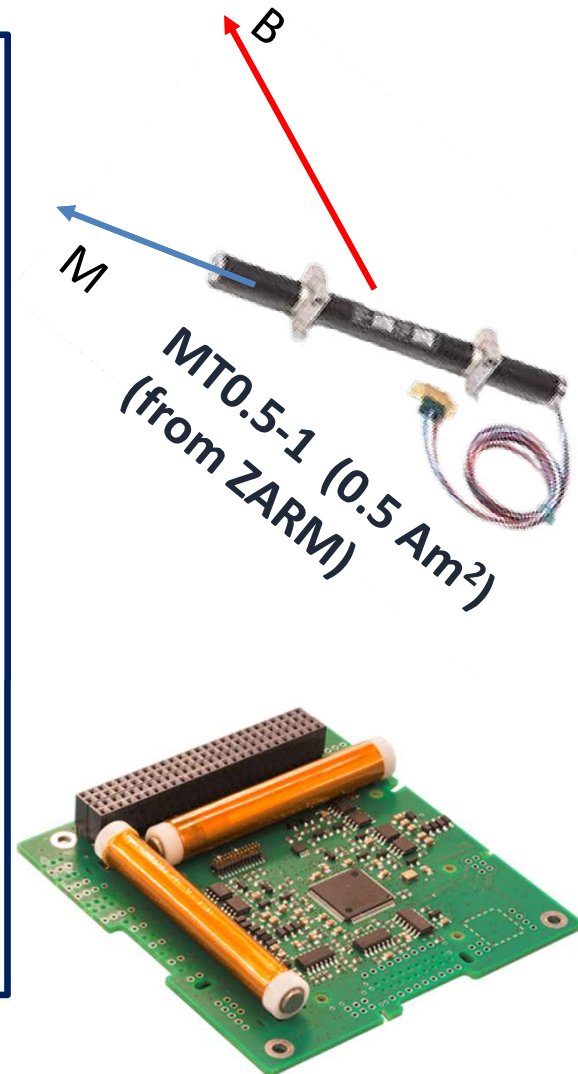


Actuators for Attitude Control

- Magnetorquers
- Reaction wheels.
- Thrusters
- Control Moment Gyros (CMG)

Torquers

- Torquers generate a dipole (M) when current flows thro the coil.
- This interacts with the magnetic field (B) and produces a torque $T_M = M \times B$
- Torquers are used for initial **detumbling** operations as well as **momentum dumping** during normal phase.
- Advantages: They are 'fuel free', 'no moving parts'
- Limitation: Torque is produced only in a plane normal to B



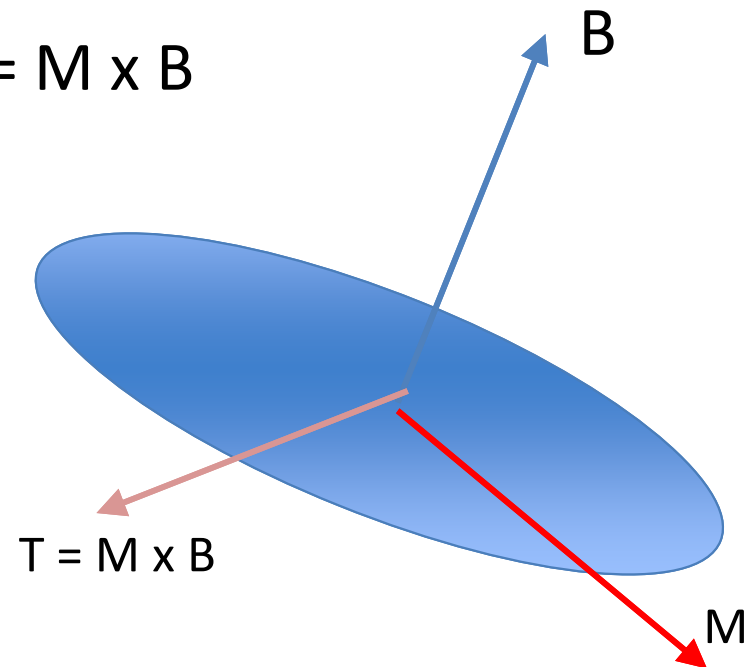
Torque by Torquers

At 500 km altitude, assuming mid-latitudes, the magnetic field (B) is about 30-40 μT .

The magnetic torque is given by $T = M \times B$

The maximum torque will be MB

M (Am^2)	Max Mag.Torque (Let $B = 30 \mu\text{T}$)
0.1	$3 \times 10^{-6} \text{ Nm}$
2.0	$6 \times 10^{-5} \text{ Nm}$
10.0	$3 \times 10^{-4} \text{ Nm}$

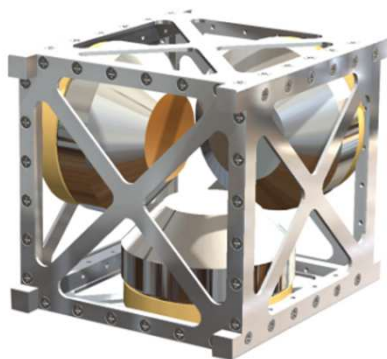


Reaction Wheels

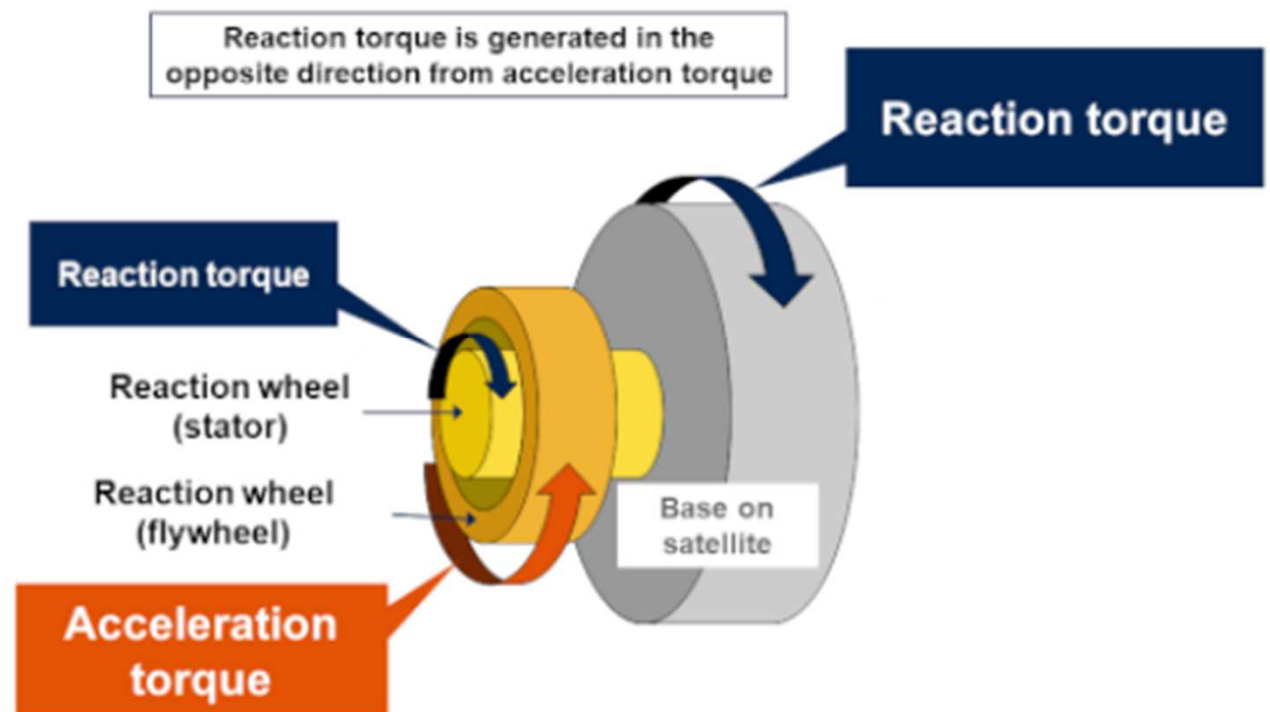
- When the wheel generates a torque in one direction, the satellite rotates in the opposite direction. This control the orientation of the satellite



Tetrahedral



Orthogonal



Reaction wheel (Torque, Momentum Equations)

Reaction wheel Torque (Nm) $T = J \frac{d\omega}{dt}$; J = Wheel Inertia

Reaction wheel Momentum (Nms) $h = J\omega$

Maximum wheel momentum is governed by the wheel inertia and the maximum speed that can be attained



Reaction Wheel



MW200

Size	100 mm (L) x 90 mm (W) x 90 mm (H)	
Mass	< 0.94 kg	
Speed Range	+/-10,000 RPM	
Angular Momentum Capacity	0.18 Nms	
Torque Capability	30 mNm	
Data Interface	Asynchronous RS-422/485	

$$Tdt = Jd\omega = dh$$

Starting from rest, if the wheel operates at a torque of 20mNm, it will saturate in 9 secs

Thrusters & CMGs

- Used for large satellites
- To increase the agility of the satellite
- Thrusters are Precise but fuel-limited.
- CMGs are best suited for large, agile, and power-conscious spacecraft where high torque is essential. For smaller, cost-sensitive missions, reaction wheels remain the more practical choice.

Attitude Control

- Why?
 - External and internal perturbation torques continuously disturb the pointing
 - For precise imaging and communication the payload need to be oriented correctly.
 - For satellite's health, power and thermal conditions are to be maintained

Attitude Dynamics (Basic)

Torque Equation in Inertial Reference Frame

$$\bar{T} = \frac{d}{dt}(\bar{h}) \quad \begin{array}{l} h \text{ is the angular momentum,} \\ T \text{ is the torque} \end{array}$$

Torque Equation in Body Reference Frame

$$\bar{T} = I\dot{\bar{\omega}} + \bar{\omega} \times I\bar{\omega} \quad \dot{\omega}_x = \frac{I_y - I_z}{I_x} \omega_y \omega_z + \frac{T_x}{I_x}$$

Expressing using Principal Inertia Reference Frame $\rightarrow$,

$$\dot{\omega}_y = \frac{I_z - I_x}{I_y} \omega_z \omega_x + \frac{T_y}{I_y}$$

T_x, T_y, T_z , are calculated by the control algorithms

$$\dot{\omega}_z = \frac{I_x - I_y}{I_z} \omega_x \omega_y + \frac{T_z}{I_z}$$

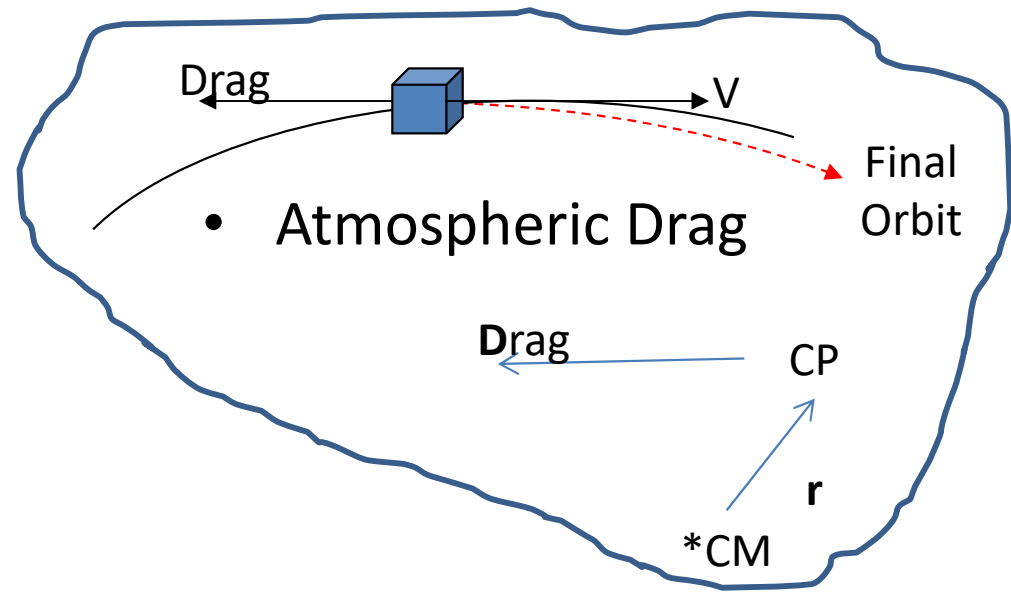
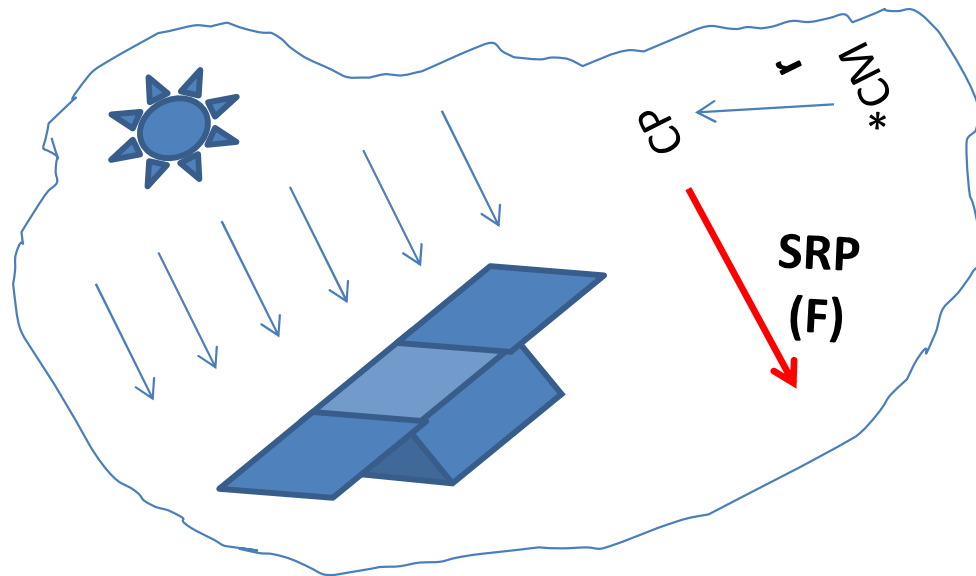
$$\bar{T} = \bar{T}_c + \bar{T}_d = I\dot{\bar{\omega}} + \bar{\omega} \times I\bar{\omega}$$

Control Torque

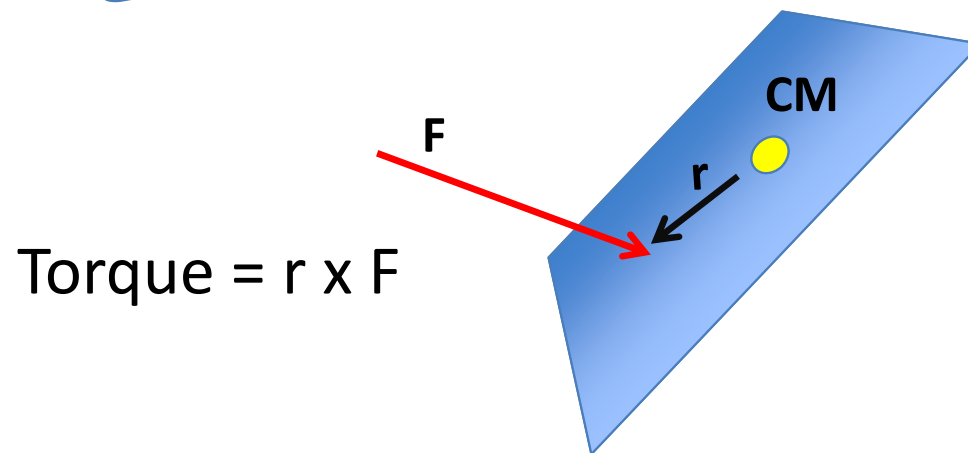
Disturbance Torque

Attitude Perturbation Sources (1/2)

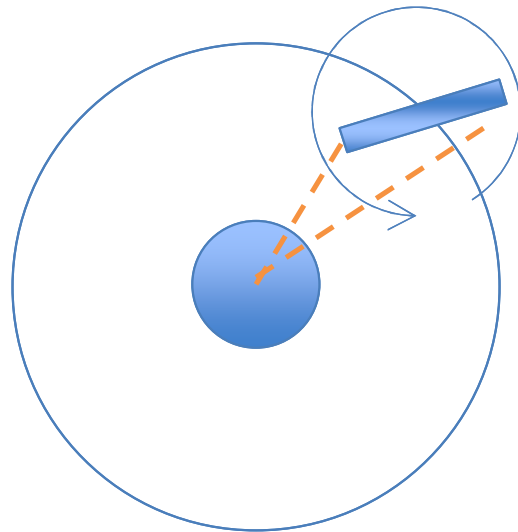
- Solar Radiation Pressure



- Atmospheric Drag

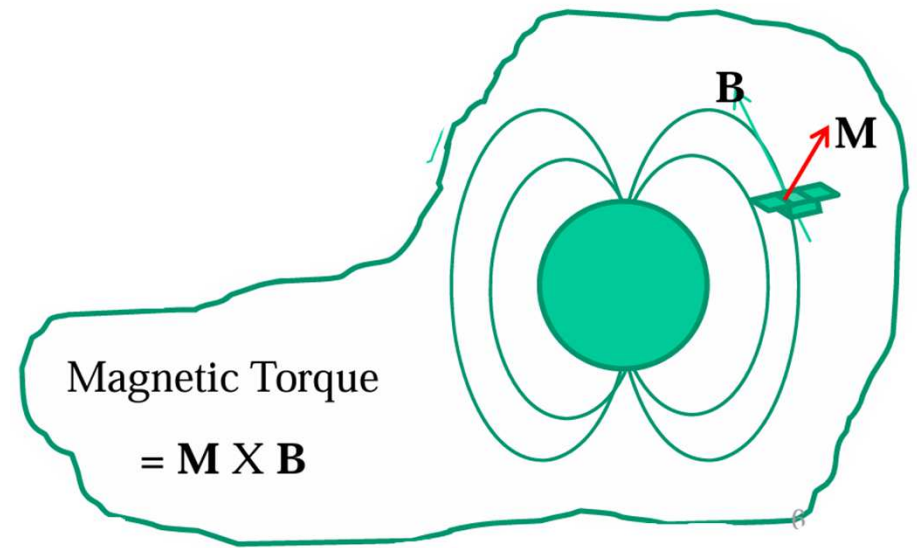
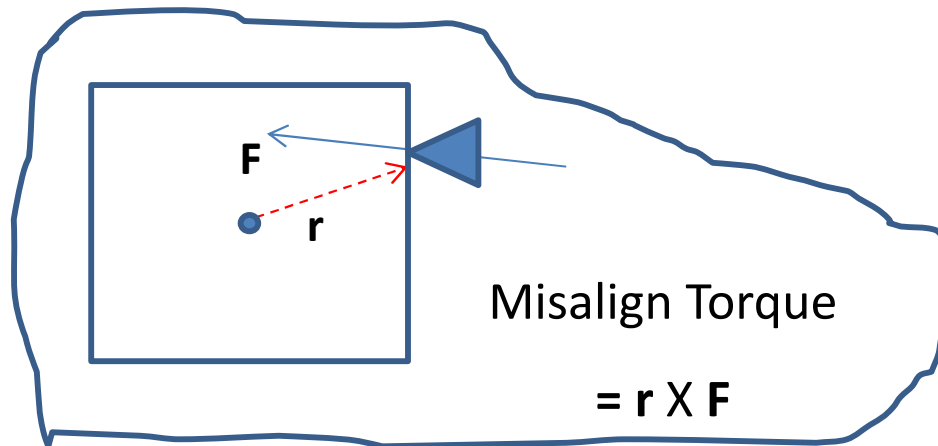


Perturbation Sources (2/2)



Gravity Gradient

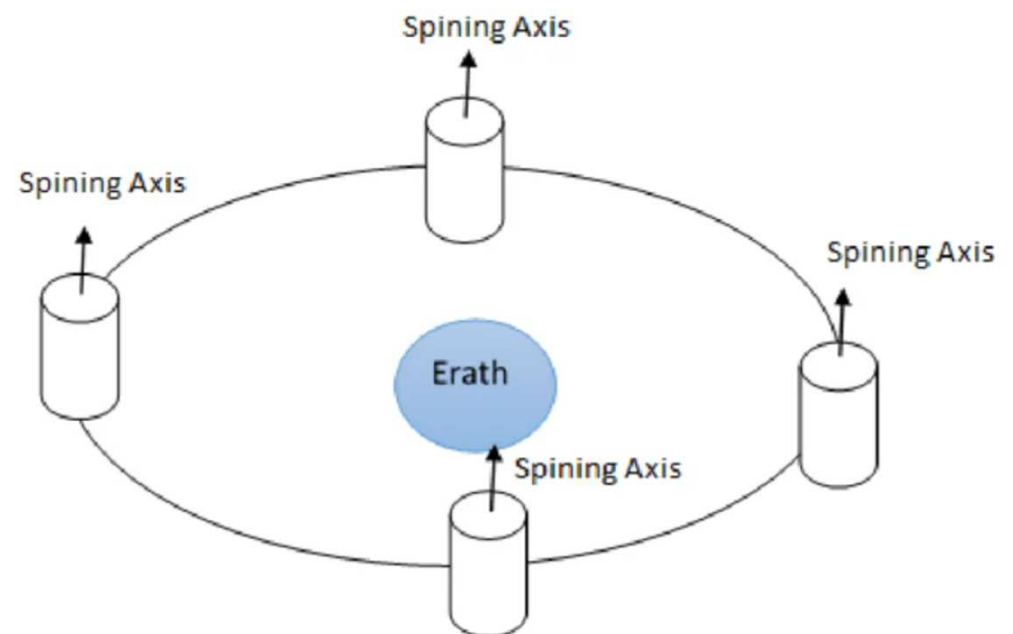
Thruster Misalignments



Residual Dipole

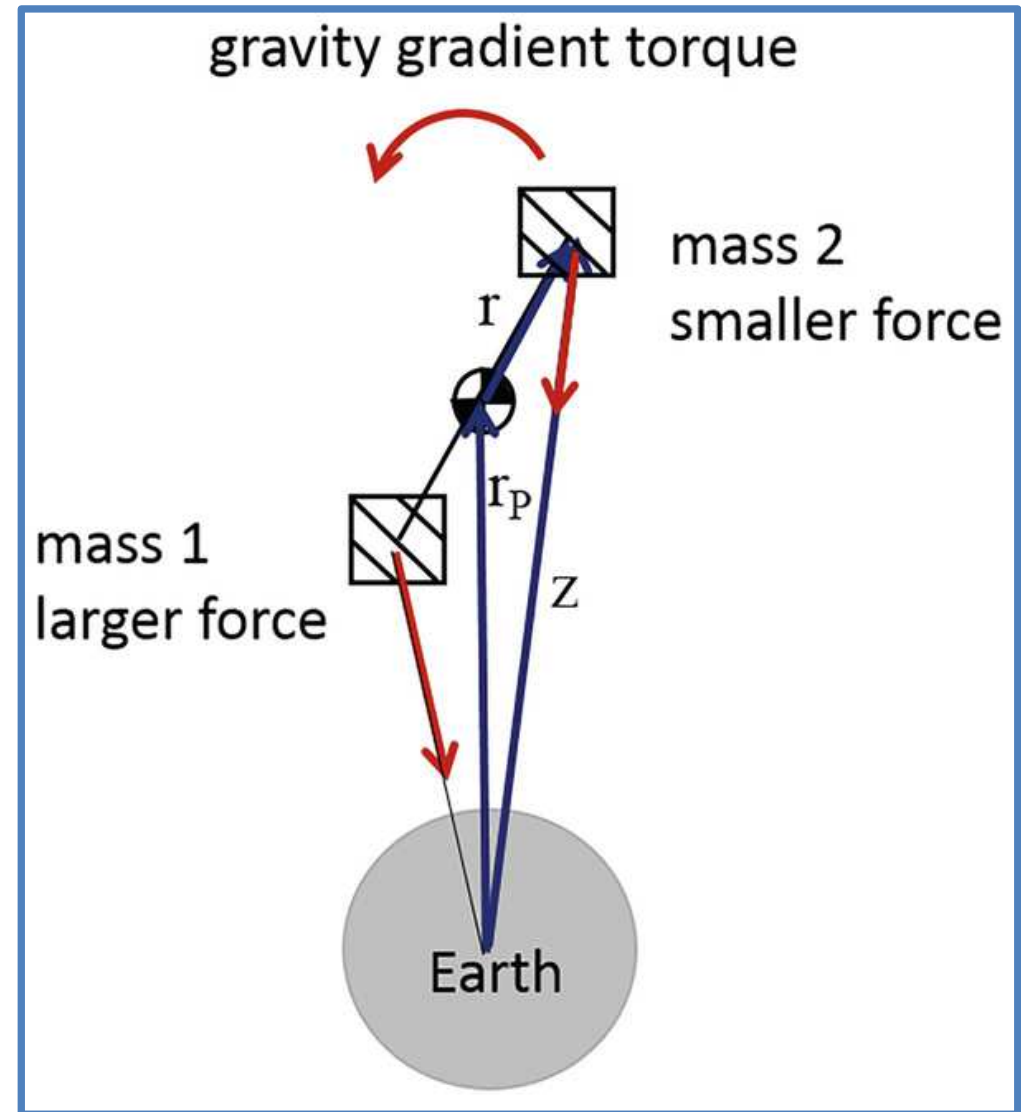
Passive Attitude Stabilization

- **Spin Stabilization:** The spin axis is inertially stable due to gyroscopic stiffness and thus maintains the direction.
- External disturbances slowly causes precession and nutation.
- As the name implies, it is passive technique; easy to implement; doesn't require complex control strategies.;

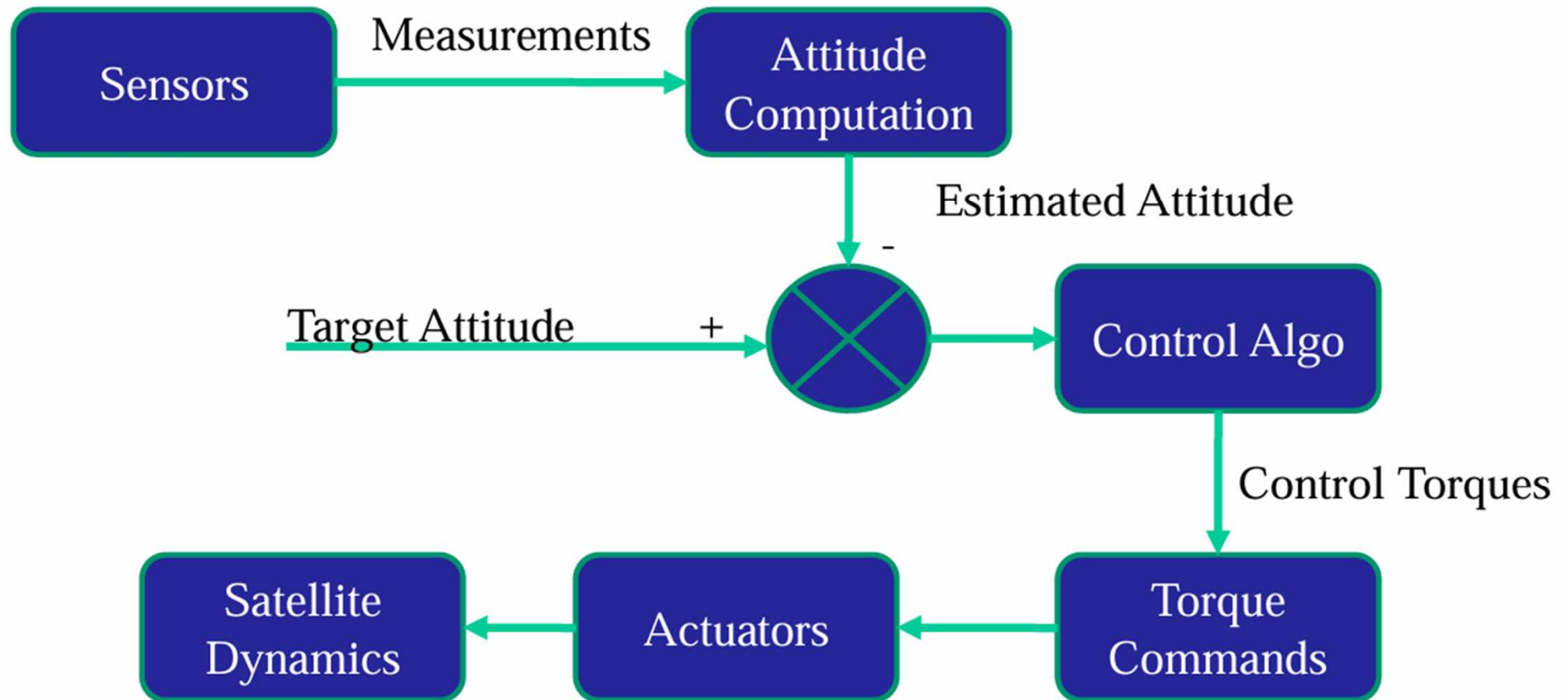


Passive Attitude Stabilization

- **Gravity Gradient Stabilization:**
For satellites which have 'pencil like' shape, the gravity gradient provides a passive stabilization. By this, the long slender axis points towards the Earth, as the satellite moves.
- As the name implies, it is passive technique; easy to implement; doesn't require complex control strategies.;
- Disturbances, causes oscillations about the radial direction; Hence a damper is needed



Attitude Determination and Control System

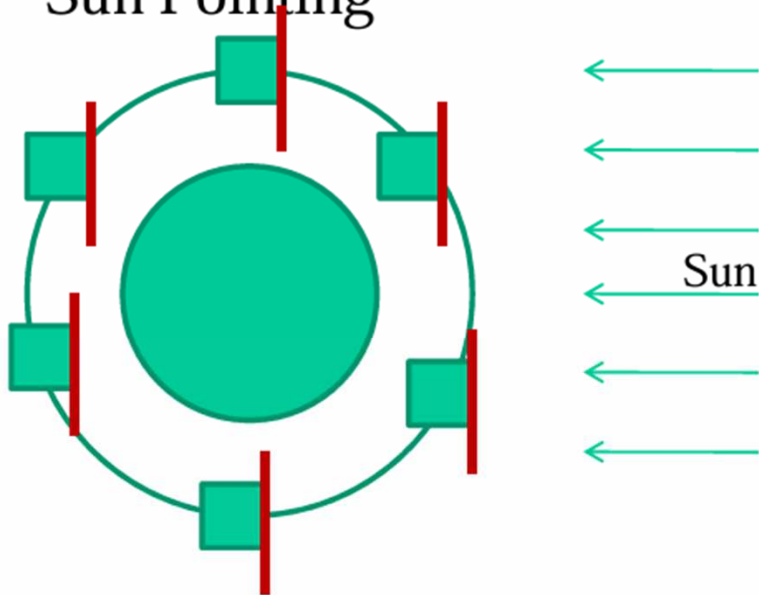


Common ADCS Modes

Modes	Remarks
Detumbling mode	To reduce the body rates due to separation disturbance
Sun Acquisition	Orient one axis towards sun for power balance (before panel deployment)
Sun Pointing	Orient solar panel towards Sun and maintain the other two axes
Earth Pointing	Orient the payload (camera, antenna) towards Earth and maintain the other two axes
Maneuver Mode	Orient the reference axis (thruster) along the DV direction and maintain the axes
Inertial Pointing	Orient a reference axis towards inertial source (ex star) and maintain the other two axes

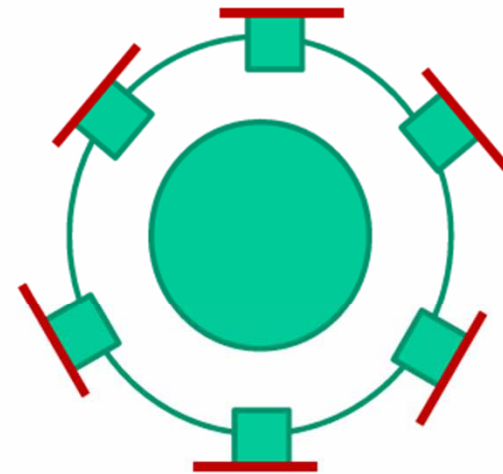
ADCS Modes

Sun Pointing

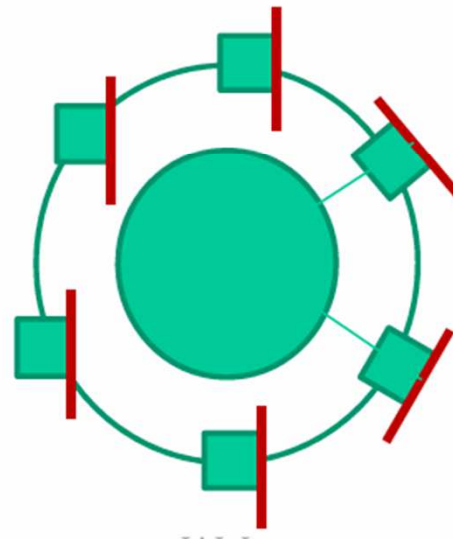


Orienting solar panel to sun for maximum power generation

Earth Pointing



For imaging and communications

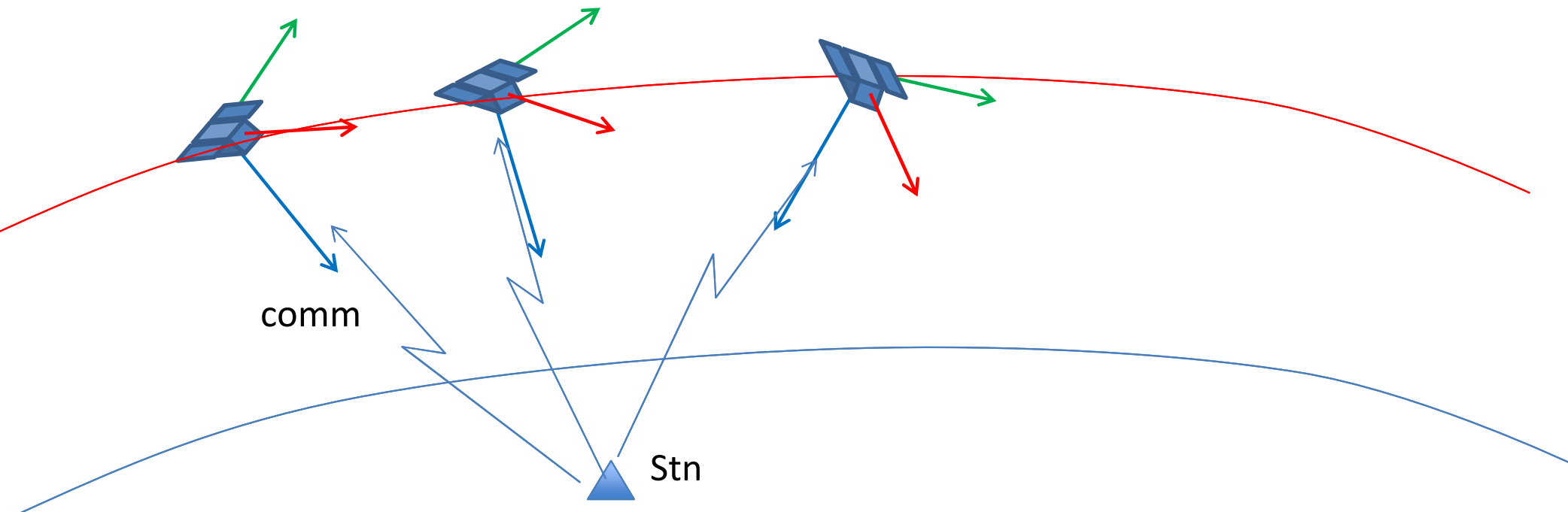


During the mission both may be required

ADCS Modes

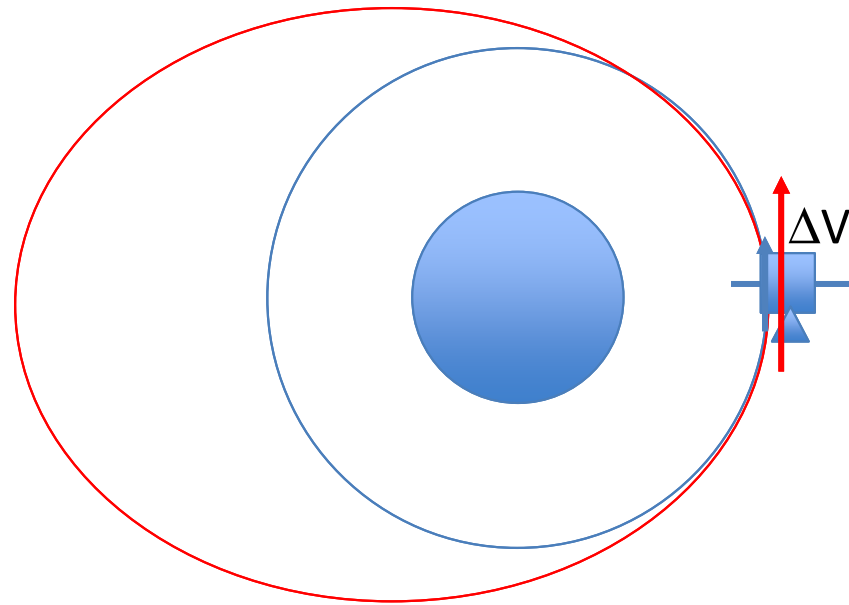
Target Pointing Mode

A reference in the body frame (say antenna axis), kept pointed towards a target (Ground station)

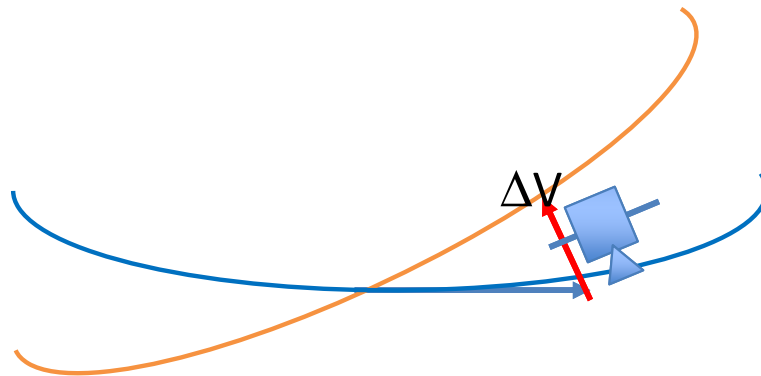


ADCS Modes

Maneuver Mode



In-Plane Maneuver



Out-of-Plane Maneuver

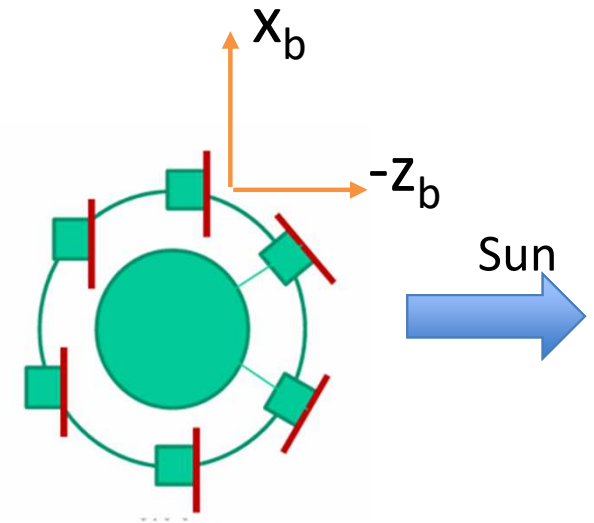
Control Algorithm

Sun Tracking Mode

Required attitude $-z_b = \text{Sun}$

$$x_b = \text{OrbNormal} \times \text{Sun}$$

$$y_b = z_b \times x_b$$



- Convert x, y, z to attitude angles or quaternion.
- Compare with the estimated current attitude
- Generate the torque command based on the errors in angle, rates
- Command the torque (T_c) to the actuators

$$\bar{T} = \bar{T}_c + \bar{T}_d = I\dot{\bar{\omega}} + \bar{\omega} \times \bar{h}$$

Control Torque

Disturbance
Torque

Control Algorithm

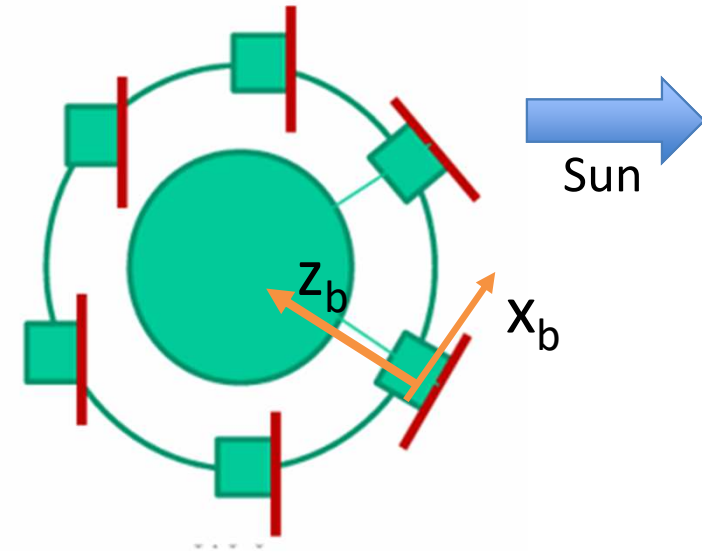
Nadir Pointing Mode

Required attitude

$$z_b = - \text{radius vector}$$

$$x_b = \text{OrbNormal} \times \text{Radius vector}$$

$$y_b = z_b \times x_b$$



- Convert x,y,z to attitude angles or quaternion.
- Compare with the estimated current attitude
- Generate the torque command based on the errors in angle, rates
- Command the torque (T_c) to the actuators

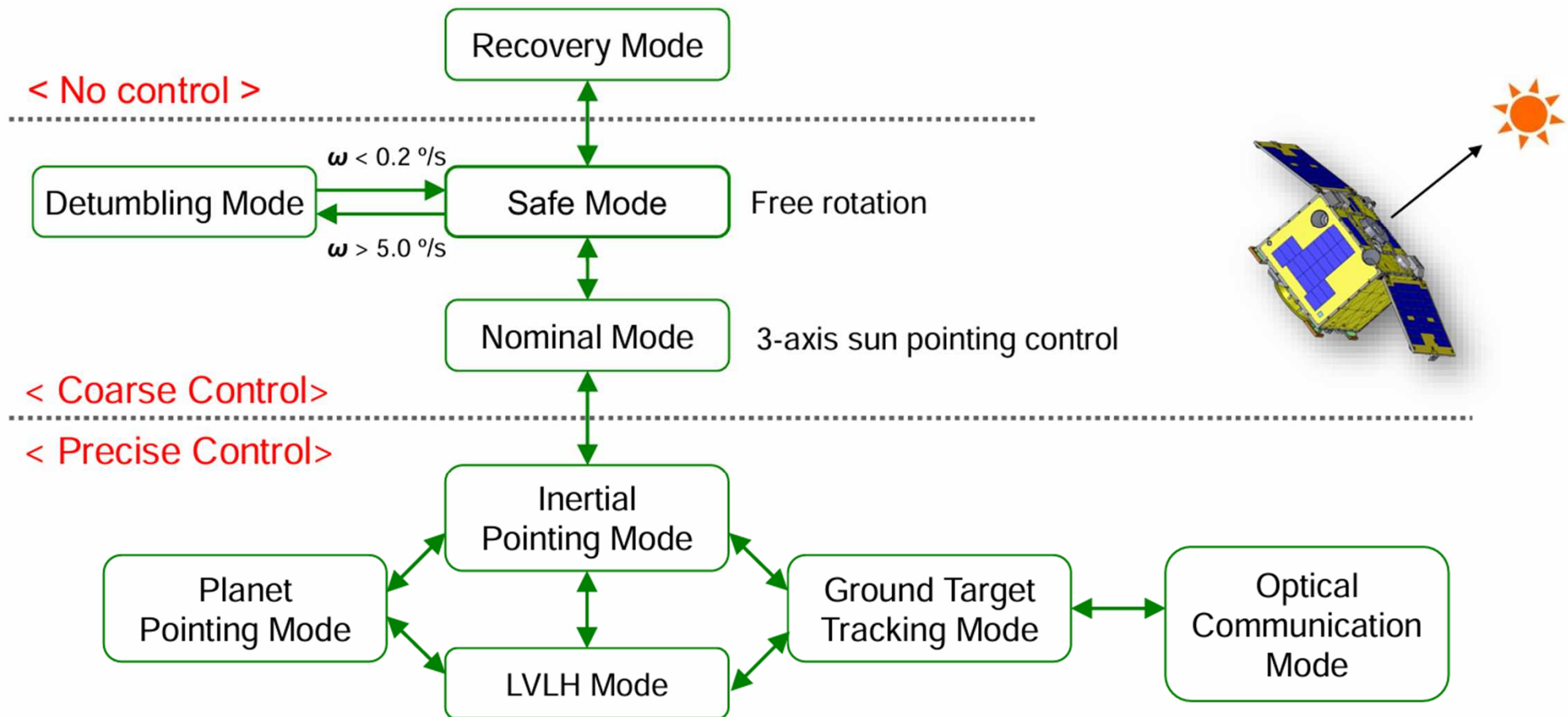
$$\bar{T} = \bar{T}_c + \bar{T}_d = I\dot{\omega} + \bar{\omega} \times \bar{h}$$

Control Torque

Disturbance
Torque

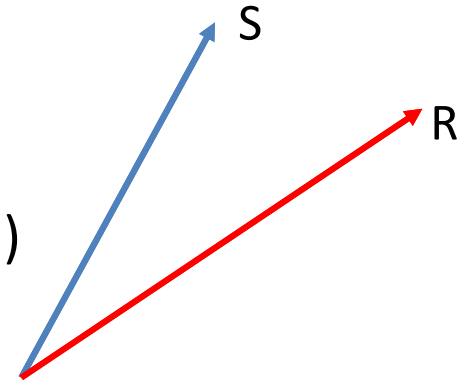
Typical State Machine for Mode Transition

During a mission, a satellite has to be operated frequently in different modes to satisfy the mission, payload requirements



Attitude Determination (Geometrical Method)

Let S and R are unit vectors measured in body frame (S_M, R_M) and also known in reference frame (S_R, R_R) thro ephemeris.



S and R could be Sun, Magnetometer measurements. If A defines the attitude matrix (DCM) of the satellite, then,

$$S_M = A S_R ; R_M = A R_R ; \text{ Define } P = \frac{R \times S}{|R \times S|} ; Q = P \times R$$

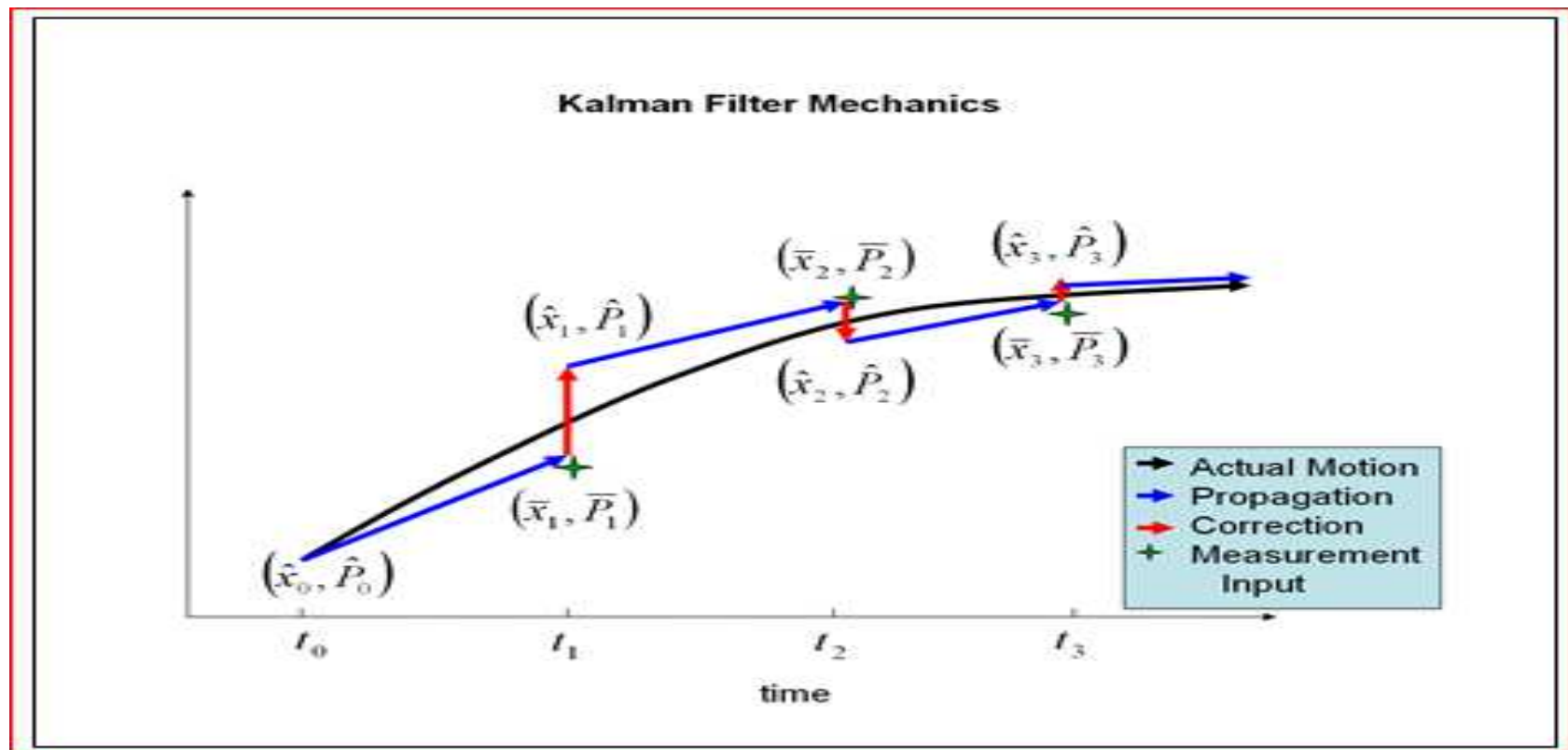
$$\text{Then } [R \quad P \quad Q]_M = A [R \quad P \quad Q]_R$$

$$\text{Then } [R \quad P \quad Q]_M [R \quad P \quad Q]_R^T = A = \text{Attitude matrix of the body.}$$

Convert A to quaternion or Euler Angles and input to control

Kalman Filter Based Attitude Determination

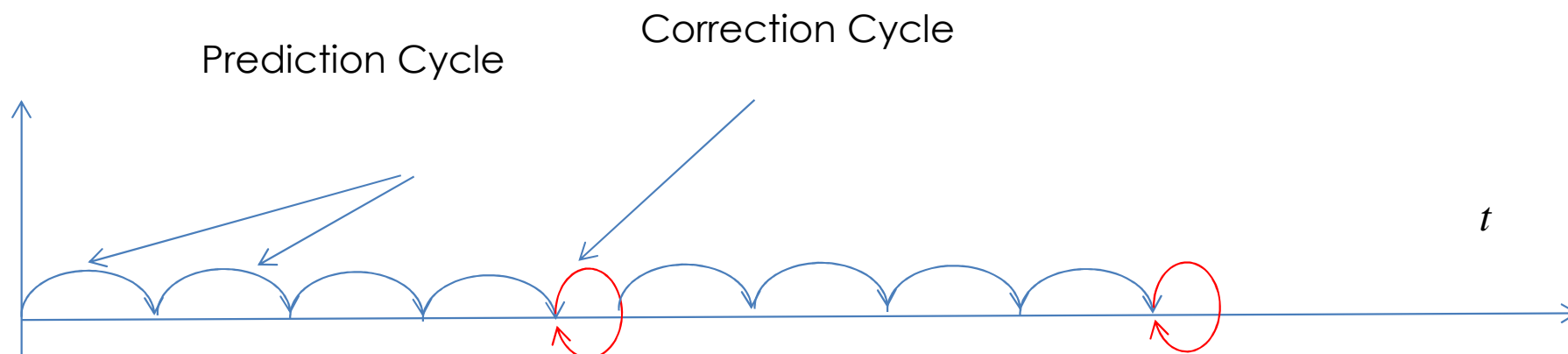
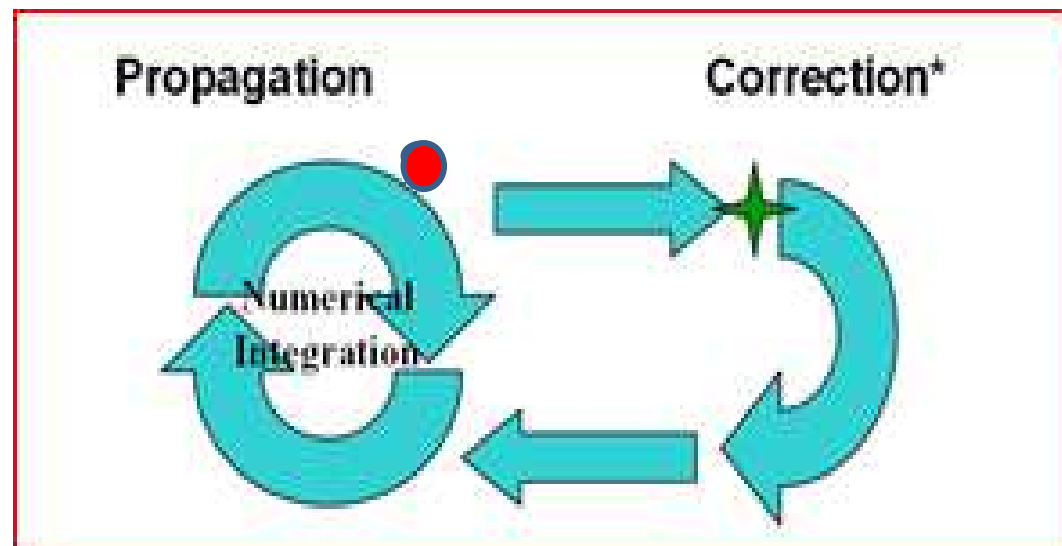
Objective is to estimate the state x (angle, rates) of a time dependent process using the measurement z which is corrupted by noise



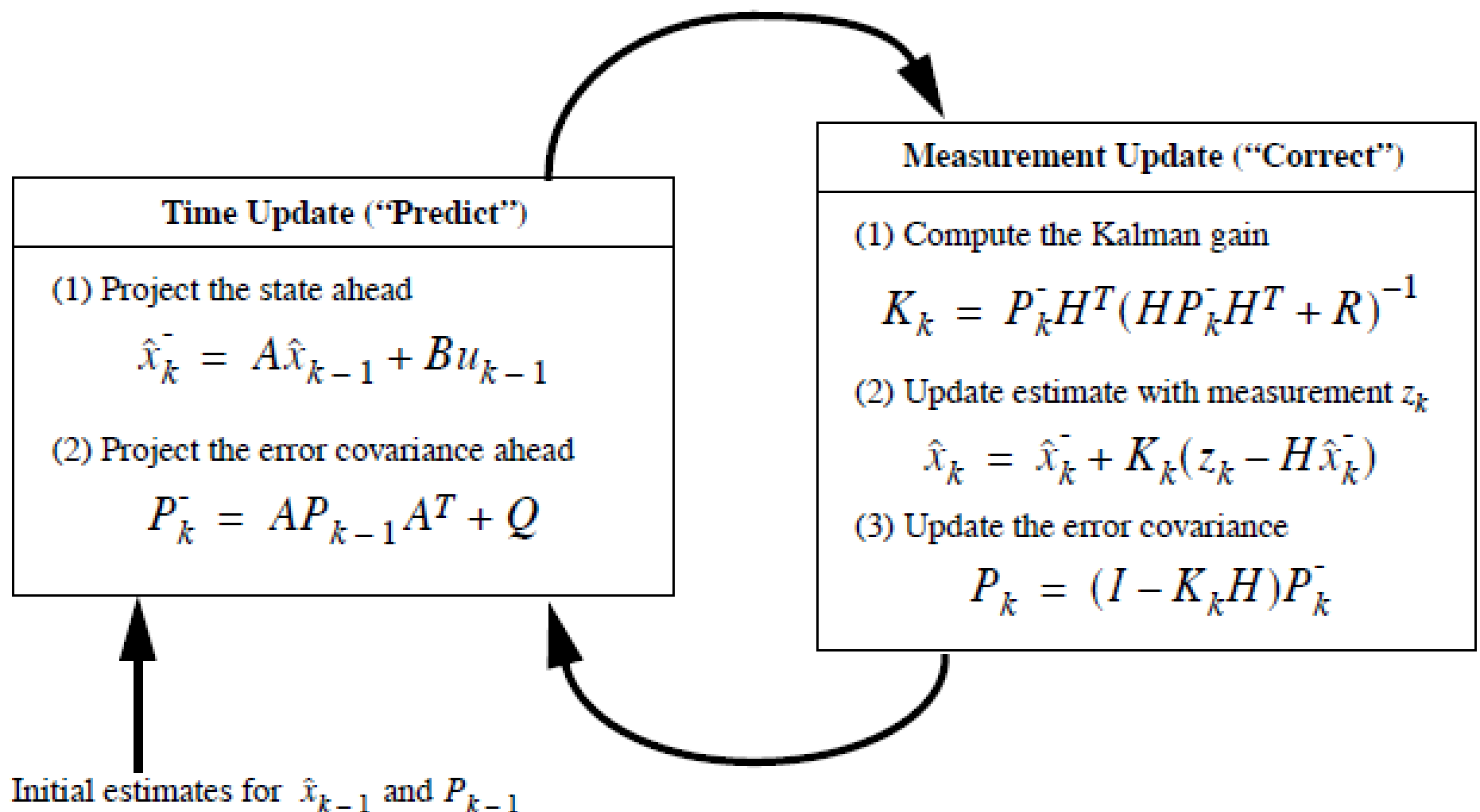
Kalman Filter Flow Diagram

Assume that propagation is at one sec interval and measurements come at 5 sec interval

Therefore, 'propagation' keeps going for 4 cycles and in the 5th cycle, when measurement is available, 'correction' cycle is executed once and again 'propagation cycle' for 4 cycles.



Kalman Filter Flow Diagram with Equations



Animation Clips

[Detumble Mode](#)

[Sun Tracking Mode](#)

[Imaging Mode](#)

Reaction Wheel Demo

https://www.youtube.com/watch?v=_UnAryf8r5Q

Case Study: CubeSat ADCS

3U CubeSat Orbit 500km alt (r=6878km)

Parameter	Value
Length (a)	0.10 m
Width (b)	0.10 m
Height (c)	0.34 m
Mass (m)	4 kg

Axis	Mol(kg·m ²)
Longitudinal (z)	~0.007
Transverse (x)	~0.041
Transverse (y)	~0.041

Gravity Gradient Torque $T \approx \frac{3\mu}{2r^3} (I_{\max} - I_{\min}) \sin(2\theta)$ $2 \times 10^{-7} \text{ Nm}$

Residual Dipole Torque: Assuming a magnetic field (B) of 30mT and a residual magnet (M) of 0.1 Am², the torque will be $MB = 3 \times 10^{-6} \text{ Nm}$

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Aerodynamic torque Torque: $v = 7.61 \text{ km/s}$; At 500 Km, density $\rho = 1 \times 10^{-11} \text{ kg/m}^3$;
cross sec area $= 0.034 \text{ m}^2$, $C_d = 2.2$,

$$F_d = \frac{1}{2} \rho v^2 C_d A = 2 \times 10^{-5} \text{ N}$$

Assuming cp to cm offset of 5cm, the torque $\sim 10^{-6} \text{ Nm}$

Challenges in ADCS

- Space environment disturbances
 - External and Internal disturbances, alignment issues, radiation effects, thermal issues...
- Limited power and mass
 - Eclipse management, seasonal variation in sun aspect angle, battery capacity, cold/hot redundancy policy,
- Sensor noise.
 - Onboard data editing for wild measurements, EKF, appropriate data rate for measurements, periodic calibration

AI for ADCS

Application	AI Contribution	Benefit
Sensor fusion	ML corrects nonlinear sensor errors	Higher attitude accuracy
Adaptive control	RL designs robust controllers	Better disturbance rejection
Fault detection	Predictive monitoring	Increased reliability
Disturbance prediction	AI forecasts drag, gravity gradient	Reduced actuator load
Mode switching	Autonomous decision-making	Less ground intervention
Actuator optimization	Smart scheduling	Longer mission life

ML = Machine Learning; RL = Reinforcement Learning

Summary Slide

- Attitude determination = knowing orientation.
- Attitude control = changing orientation.
- Sensors + actuators + algorithms = ADCS.
- Coordinate Systems, Transformation
- Perturbations in space and typical perturbations on a cubesat

Discussion / Q&A